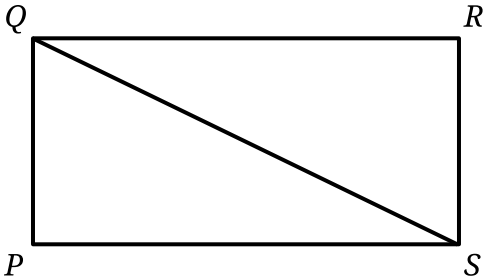


Question ID: 2c3aefc9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In rectangle $PQRS$, the length of line segment QS is $\sqrt{2,344}$ and the length of side PS is 8 more than the length of side RS . What is the length of side QR ?

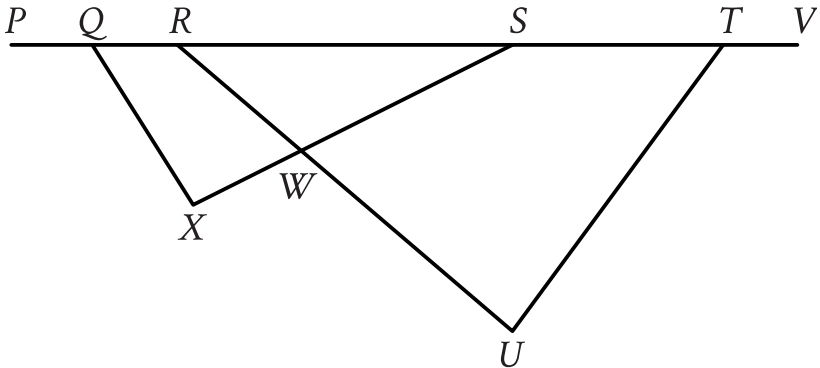
Correct Answer: 38

Rationale

The correct answer is 38. It's given that in rectangle $PQRS$, the length of side PS is 8 more than the length of side RS . In a rectangle, opposite sides are congruent. Since side QR is opposite side PS , it follows that side QR is congruent to side PS . Thus, the length of side QR is 8 more than the length of side RS . Let x represent the length of side RS . It follows that the length of side QR is represented by the expression $x + 8$. Since all angles in a rectangle are right angles, angle R is a right angle. It follows that side QR , side RS , and line segment QS form a right triangle, where line segment QS is the hypotenuse and sides QR and RS are the legs. It's given that the length of line segment QS is $\sqrt{2,344}$. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting x for a , $x + 8$ for b , and $\sqrt{2,344}$ for c in this equation yields $x^2 + (x + 8)^2 = (\sqrt{2,344})^2$, or $x^2 + (x + 8)^2 = 2,344$. Expanding the left-hand side of this equation yields $x^2 + x^2 + 16x + 64 = 2,344$, or $2x^2 + 16x + 64 = 2,344$. Dividing both sides of this equation by 2 yields $x^2 + 8x + 32 = 1,172$. To complete the square, subtracting 16 from both sides of this equation yields $x^2 + 8x + 16 = 1,156$. Since the left-hand side of this equation is a perfect square trinomial, it follows that $x^2 + 8x + 16 = 1,156$ is equivalent to $(x + 4)^2 = 1,156$. Taking the positive square root of both sides of this equation yields $x + 4 = 34$. Subtracting 4 from both sides of this equation yields $x = 30$. Since the length of side QR is represented by the expression $x + 8$, it follows that the length of side QR is $30 + 8$, or 38.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, points Q , R , S , and T lie on line segment PV , and line segment RU intersects line segment SX at point W . The measure of $\angle SQX$ is 48° , the measure of $\angle SXQ$ is 86° , the measure of $\angle SWU$ is 85° , and the measure of $\angle VTU$ is 162° . What is the measure, in degrees, of $\angle TUR$?

Correct Answer: 123

Rationale

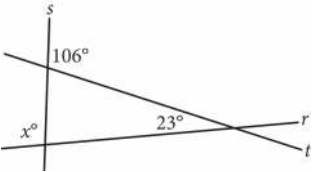
The correct answer is **123**. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is **180** degrees. It's given that the measure of $\angle SQX$ is 48° and the measure of $\angle SXQ$ is 86° . Since points S , Q , and X form a triangle, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle QSX$ is $180 - 48 - 86$, or **46**. It's also given that the measure of $\angle SWU$ is 85° . Since $\angle SWU$ and $\angle SWR$ are supplementary angles, the sum of their measures is **180** degrees. It follows that the measure, in degrees, of $\angle SWR$ is $180 - 85$, or **95**. Since points R , S , and W form a triangle, and $\angle RSW$ is the same angle as $\angle QSX$, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle WRS$ is $180 - 46 - 95$, or **39**. It's given that the measure of $\angle VTU$ is 162° . Since $\angle VTU$ and $\angle STU$ are supplementary angles, the sum of their measures is **180** degrees. It follows that the measure, in degrees, of $\angle STU$ is $180 - 162$, or **18**. Since points R , T , and U form a triangle, and $\angle URT$ is the same angle as $\angle WRS$, it follows from the triangle angle sum theorem that the measure, in degrees, of $\angle TUR$ is $180 - 39 - 18$, or **123**.

Question ID: f88f27e5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Intersecting lines r, s, and t are shown below.



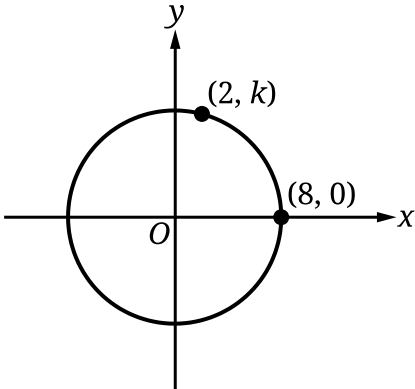
What is the value of x ?

Rationale

The correct answer is 97. The intersecting lines form a triangle, and the angle with measure of x° is an exterior angle of this triangle. The measure of an exterior angle of a triangle is equal to the sum of the measures of the two nonadjacent interior angles of the triangle. One of these angles has measure of 23° and the other, which is supplementary to the angle with measure 106° , has measure of $180^\circ - 106^\circ = 74^\circ$. Therefore, the value of x is $23 + 74 = 97$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Note: Figure not drawn to scale.

The circle shown has its center at $(0, 0)$. What is the value of k ?

Answer

- A. 6
- B. 7
- C. 8
- D. $\sqrt{60}$

Correct Answer: D

Rationale

Choice D is correct. It's given that the circle shown has its center at $(0, 0)$. The equation of a circle in the xy -plane with its center at $(0, 0)$ can be expressed as $x^2 + y^2 = r^2$, where r is the radius of the circle. It's shown that $(8, 0)$ lies on the circle. Since the circle has its center at $(0, 0)$ and $(8, 0)$ lies on the circle, it follows that the radius of the circle is 8. Substituting 8 for r in the equation $x^2 + y^2 = r^2$ yields $x^2 + y^2 = 64$. It's also shown that $(2, k)$ lies on the circle. Substituting 2 for x and k for y in this equation yields $2^2 + k^2 = 64$, or $4 + k^2 = 64$. Subtracting 4 from both sides of this equation yields $k^2 = 60$. Taking the square root of both sides of this equation yields $k = \pm\sqrt{60}$. Since $(2, k)$ lies above the x -axis, k must be positive. Therefore $k = \sqrt{60}$.

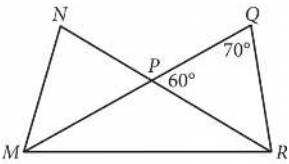
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the x -coordinate of one of the circle's x -intercepts, not the value of k .

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, $\overline{MQ}$ and $\overline{NR}$ intersect at point P, $NP = QP$, and $MP = PR$. What is the measure, in degrees, of $\angle QMR$? (Disregard the degree symbol when gridding your answer.)

Rationale

The correct answer is 30. It is given that the measure of $\angle QPR$ is 60° . Angle MPR and $\angle QPR$ are collinear and therefore are supplementary angles. This means that the sum of the two angle measures is 180° , and so the measure of $\angle MPR$ is 120° . The sum of the angles in a triangle is 180° . Subtracting the measure of $\angle MPR$ from 180° yields the sum of the other angles in the triangle MPR. Since $180 - 120 = 60$, the sum of the measures of $\angle QMR$ and $\angle NRM$ is 60° . It is given that $MP = PR$, so it follows that triangle MPR is isosceles. Therefore $\angle QMR$ and $\angle NRM$ must be congruent. Since the sum of the measure of these two angles is 60° , it follows that the measure of each angle is 30° .

An alternate approach would be to use the exterior angle theorem, noting that the measure of $\angle QPR$ is equal to the sum of the measures of $\angle QMR$ and $\angle NRM$. Since both angles are equal, each of them has a measure of 30° .

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

Answer

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

Correct Answer: C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 meters. Therefore, $h = 6$. It's also given that the right circular cylinder has a base with a circumference of 20π meters. The circumference, C , of a circle is given by $C = 2\pi r$, where r is the radius of the circle. Substituting 20π for C in the formula $C = 2\pi r$ yields $20\pi = 2\pi r$. Dividing each side of this equation by 2π yields $10 = r$. Substituting 10 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi(10)^2(6)$, or $V = 600\pi$. Therefore, the volume, in cubic meters, of the cylinder is 600π .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the lateral surface area, not the volume, of the cylinder.

Choice D is incorrect. This is the result of using the diameter, not the radius, for the value of r in the formula $V = \pi r^2 h$.

Question ID: 1fe41c6b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The table gives the areas and perimeters of two similar rectangles, where n is a constant.

	Area (square inches)	Perimeter (inches)
Rectangle A	630	210
Rectangle B	2,520	n

What is the value of n ?

Answer

- A. 2,100
- B. 1,680
- C. 840
- D. 420

Correct Answer: D

Rationale

Choice D is correct. It’s given that rectangle A and rectangle B are similar. Therefore, there is a positive constant k such that the length of each side of rectangle B is k times the length of the corresponding side of rectangle A and the area of rectangle B is k^2 times the area of rectangle A. It’s also given that the area of rectangle A is 630 square inches and that the area of rectangle B is 2,520 square inches. It follows that $2,520 = 630k^2$. Dividing both sides of this equation by 630 yields $4 = k^2$, or $k^2 = 4$. Taking the positive square root of each side of this equation yields $k = 2$. Therefore, the length of each side of rectangle B is 2 times the length of the corresponding side of rectangle A. It follows that the perimeter of rectangle B is 2 times the perimeter of rectangle A. Since the perimeter of rectangle A is 210 inches, the perimeter of rectangle B is $2(210)$, or 420, inches. Therefore, $n = 420$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from using the area scale factor instead of the side length scale factor.

Question ID: 27d075d8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 52$ and $4x + 13$, respectively. What is the value of x ?

Answer

- A. 5
- B. 13
- C. 23
- D. 38

Correct Answer: A

Rationale

Choice A is correct. It’s given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other, the angles are complementary. Therefore, the sum of the measures, in degrees, of $\angle Q$ and $\angle R$ is 90. It’s given that the measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 52$ and $4x + 13$, respectively. Therefore, $(x + 52) + (4x + 13) = 90$. Combining like terms in this equation yields $5x + 65 = 90$. Subtracting 65 from each side of this equation results in $5x = 25$. Dividing each side of this equation by 5 yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\angle Q$ and $\angle R$ were congruent rather than complementary.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

Answer

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

Correct Answer: C

Rationale

Choice C is correct. It's given that the measure of angle A is 52° and the measure of angle P is 52° . Therefore, angle A is congruent to angle P . Choice C states that the measures of angle B and angle R are 32° and 96° , respectively. The triangle sum theorem states that the measures of the angles in a triangle sum to 180° . In triangle ABC , if the measure of angle B is 32° , it follows that the measure of angle C is $(180 - (52 + 32))^\circ$, or 96° . In triangle PQR , if the measure of angle R is 96° , it follows that the measure of angle Q is $(180 - (52 + 96))^\circ$, or 32° . Therefore, in triangles ABC and PQR , angle B is congruent to angle Q , and angle C is congruent to angle R . If two triangles have three pairs of congruent angles, then the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle ABC is similar to triangle PQR is that the measures of angle B and angle R are 32° and 96° , respectively.

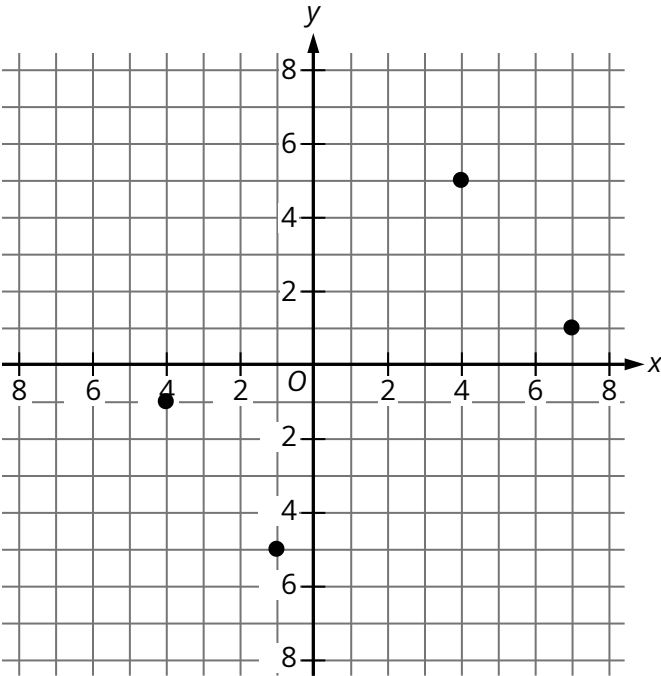
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. If the measure of angle B is 52° , it follows that the measure of angle C is $(180 - (52 + 52))^\circ$, or 76° . If the measure of angle Q is 32° , which isn't equal to any angle measure in triangle ABC , the triangles can't be similar.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Correct Answer: 50

Rationale

The correct answer is 50. The area of a rectangle is **Math output error**, where **Math output error** and w are the lengths of two adjacent sides of the rectangle. It's given that the points $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$ form a rectangle. The points $(-4, -1)$ and $(4, 5)$ form a side of the rectangle that is adjacent to the side of the rectangle formed by the points $(4, 5)$ and $(7, 1)$. In the xy -plane, the distance between two points, (x_1, y_1) and (x_2, y_2) , can be found as $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Thus, the length, in units, of the side formed by the points $(-4, -1)$ and $(4, 5)$ can be found as $\sqrt{(4 - (-4))^2 + (5 - (-1))^2}$, which can be simplified to $\sqrt{100}$, or 10. The length, in units, of the side formed by the points $(4, 5)$ and $(7, 1)$ can be found as $\sqrt{(7 - 4)^2 + (1 - 5)^2}$, which can be simplified to $\sqrt{25}$, or 5. Therefore, the area, in square units, of the rectangle formed by the four points shown is $(10)(5)$, or 50.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + x\sqrt{t} + y^2 - y - 65 = 0$$

The given equation, where t is a constant, defines a circle in the xy -plane. The radius of this circle is $\sqrt{83}$. What is the value of t ?

Answer

- A. 73
- B. 71
- C. 37
- D. 35

Correct Answer: B

Rationale

Choice B is correct. It's given that the equation $x^2 + x\sqrt{t} + y^2 - y - 65 = 0$, where t is a constant, defines a circle in the xy -plane. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius of the circle. Adding 65 to both sides of the given equation yields $x^2 + x\sqrt{t} + y^2 - y = 65$. By completing the square, this equation can be rewritten as $\left(x^2 + x\sqrt{t} + \left(\frac{\sqrt{t}}{2}\right)^2\right) + \left(y^2 - y + \left(-\frac{1}{2}\right)^2\right) = 65 + \left(\frac{\sqrt{t}}{2}\right)^2 + \left(-\frac{1}{2}\right)^2$. This equation can be rewritten as $\left(x^2 + x\sqrt{t} + \frac{t}{4}\right) + \left(y^2 - y + \frac{1}{4}\right) = 65 + \frac{t}{4} + \frac{1}{4}$, or $\left(x + \frac{\sqrt{t}}{2}\right)^2 + \left(y - \frac{1}{2}\right)^2 = \frac{261}{4} + \frac{t}{4}$. Therefore, $r^2 = \frac{261}{4} + \frac{t}{4}$. It's given that the radius of this circle is $\sqrt{83}$; therefore, $r = \sqrt{83}$. Squaring both sides of this equation yields $r^2 = 83$. Since $r^2 = \frac{261}{4} + \frac{t}{4}$ and $r^2 = 83$, it follows that $\frac{261}{4} + \frac{t}{4} = 83$. Multiplying both sides of this equation by 4 yields $261 + t = 332$. Subtracting 261 from both sides of this equation yields $t = 71$. Therefore, the value of t is 71.

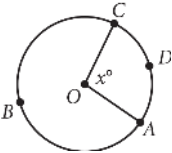
Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



The circle above has center O, the length of arc $\widehat{ADC}$ is 5π , and $x = 100$. What is the length of arc $\widehat{ABC}$?

Answer

- A. 9π
- B. 13π
- C. 18π
- D. $\frac{13}{2}\pi$

Correct Answer: B

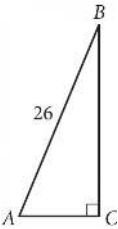
Rationale

Choice B is correct. The ratio of the lengths of two arcs of a circle is equal to the ratio of the measures of the central angles that subtend the arcs. It's given that arc $\widehat{ADC}$ is subtended by a central angle with measure 100° . Since the sum of the measures of the angles about a point is 360° , it follows that arc $\widehat{ABC}$ is subtended by a central angle with measure $360^\circ - 100^\circ = 260^\circ$. If s is the length of arc $\widehat{ABC}$, then s must satisfy the ratio $\frac{s}{5\pi} = \frac{260}{100}$. Reducing the fraction $\frac{260}{100}$ to its simplest form gives $\frac{13}{5}$. Therefore, $\frac{s}{5\pi} = \frac{13}{5}$. Multiplying both sides of $\frac{s}{5\pi} = \frac{13}{5}$ by 5π yields $s = 13\pi$.

Choice A is incorrect. This is the length of an arc consisting of exactly half of the circle, but arc $\widehat{ABC}$ is greater than half of the circle. Choice C is incorrect. This is the total circumference of the circle. Choice D is incorrect. This is half the length of arc $\widehat{ABC}$, not its full length.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

Rationale

The correct answer is 24. The sine of an acute angle in a right triangle is equal to the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the sine of angle B, or $\sin(B)$, is equal to the ratio of the length of side $\overline{AC}$ to the length of side $\overline{AB}$. It's given that the length of side $\overline{AB}$ is 26 and that $\sin(B) = \frac{5}{13}$. Therefore, $\frac{5}{13} = \frac{AB}{26}$. Multiplying both sides of this equation by 26 yields $AC = 10$.

By the Pythagorean Theorem, the relationship between the lengths of the sides of triangle ABC is as follows: $26^2 = 10^2 + BC^2$, or $676 = 100 + BC^2$. Subtracting 100 from both sides of $676 = 100 + BC^2$ yields $576 = BC^2$. Taking the square root of both sides of $576 = BC^2$ yields $24 = BC$.

Question ID: f7dbde16

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

Answer

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

Correct Answer: D

Rationale

Choice D is correct. Two triangles are similar if they have three pairs of congruent corresponding angles. It's given that angles L and R each measure 60° , and so these corresponding angles are congruent. If angle M is 70° , then angle N must be 50° so that the sum of the angles in triangle LMN is 180° . If angle T is 50° , then angle S must be 70° so that the sum of the angles in triangle RST is 180° . Therefore, if the measures of angles M and T are 70° and 50° , respectively, then corresponding angles M and S are both 70° , and corresponding angles N and T are both 50° . It follows that triangles LMN and RST have three pairs of congruent corresponding angles, and so the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle LMN is similar to triangle RST is that the measures of angles M and T are 70° and 50° , respectively.

Choice A is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the two sides given are not proportional and the angle given is not included by the given sides.

Choice B is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the angle given is not included between the proportional sides.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

The perimeter of an isosceles right triangle is $18 + 18\sqrt{2}$ inches. What is the length, in inches, of the hypotenuse of this triangle?

Answer

- A. 9
- B. $9\sqrt{2}$
- C. 18
- D. $18\sqrt{2}$

Correct Answer: C

Rationale

Choice C is correct. The perimeter of a triangle is the sum of the lengths of its sides. Since the given triangle is an isosceles right triangle, the length of each leg is the same and the length of the hypotenuse is equal to $\sqrt{2}$ times the length of a leg. Let x represent the length, in inches, of a leg of this isosceles right triangle. Therefore, the perimeter, in inches, of the triangle is $x + x + x\sqrt{2}$, or $2x + x\sqrt{2}$, which is equivalent to $x(2 + \sqrt{2})$. It's given that the perimeter of this triangle is $18 + 18\sqrt{2}$ inches. Thus, $x(2 + \sqrt{2}) = 18 + 18\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $x = \frac{18+18\sqrt{2}}{2+\sqrt{2}}$. Multiplying the right-hand side of this equation by $\frac{2-\sqrt{2}}{2-\sqrt{2}}$ yields $x = \frac{36+36\sqrt{2}-18\sqrt{2}-36}{2}$, or $x = 9\sqrt{2}$. It follows that the length, in inches, of a leg of this isosceles right triangle is $9\sqrt{2}$. Therefore, the length, in inches, of the hypotenuse of this isosceles right triangle is $(9\sqrt{2})(\sqrt{2})$, or 18.

Choice A is incorrect. If this were the length of the hypotenuse, the perimeter would be $9 + 9\sqrt{2}$ inches.

Choice B is incorrect. This is the length, in inches, of a leg of this triangle, not the hypotenuse.

Choice D is incorrect. If this were the length of the hypotenuse, the perimeter would be $36 + 18\sqrt{2}$ inches.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Point F lies on a unit circle in the xy -plane and has coordinates $(1, 0)$. Point G is the center of the circle and has coordinates $(0, 0)$. Point H also lies on the circle and has coordinates $(-1, y)$, where y is a constant. Which of the following could be the positive measure of angle FGH , in radians?

Answer

- A. $\frac{27\pi}{2}$
- B. $\frac{29\pi}{2}$
- C. 24π
- D. 25π

Correct Answer: D

Rationale

Choice D is correct. It's given that the circle is a unit circle, which means the circle has a radius of 1. It's also given that point G is the center of the circle and has coordinates $(0, 0)$ and that point H lies on the circle and has coordinates $(-1, y)$. Since the radius of the circle is 1, the value of y must be 0, as all other points with an x -coordinate of -1 are a distance greater than 1 away from point G . Since F and H are points on the unit circle centered at G , let side FG be the starting side of the angle and side GH be the terminal side of the angle. Since side FG is on the positive x -axis and side GH is on the negative x -axis, side FG is half of a rotation around the unit circle, or π radians, away from side GH . Therefore, the positive measure of angle FGH , in radians, is equal to π plus an integer multiple of 2π . In other words, the positive measure of angle FGH , in radians, is an odd integer multiple of π . Of the given choices, only 25π is an odd integer multiple of π .

Choice A is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0, -1)$, not $(-1, 0)$.

Choice B is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(0, 1)$, not $(-1, 0)$.

Choice C is incorrect. This could be the positive measure of an angle where the starting side is FG and the terminal side contains the point $(1, 0)$, not $(-1, 0)$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$x^2 + 20x + y^2 + 16y = -20$

The equation above defines a circle in the xy -plane. What are the coordinates of the center of the circle?

Answer

- A. $(-20, -16)$
- B. $(-10, -8)$
- C. $(10, 8)$
- D. $(20, 16)$

Correct Answer: B

Rationale

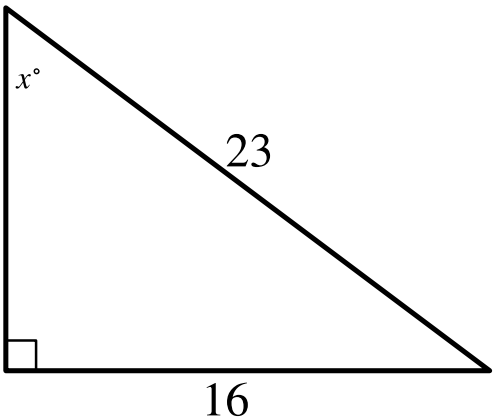
Choice B is correct. The standard equation of a circle in the xy -plane is of the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) are the coordinates of the center of the circle and r is the radius. The given equation can be rewritten in standard form by completing the squares. So the sum of the first two terms, $x^2 + 20x$, needs a 100 to complete the square, and the sum of the second two terms, $y^2 + 16y$, needs a 64 to complete the square. Adding 100 and 64 to both sides of the given equation yields $(x^2 + 20x + 100) + (y^2 + 16y + 64) = -20 + 100 + 64$, which is equivalent to $(x + 10)^2 + (y + 8)^2 = 144$. Therefore, the coordinates of the center of the circle are $(-10, -8)$.

Choices A, C, and D are incorrect and may result from computational errors made when attempting to complete the squares or when identifying the coordinates of the center.

Question ID: 1429dcd

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^\circ$?

Correct Answer: .6956, .6957, 16/23

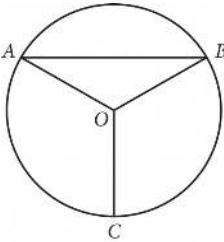
Rationale

The correct answer is $\frac{16}{23}$. In a right triangle, the sine of an acute angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the length of the side opposite the angle with measure x° is 16 units and the length of the hypotenuse is 23 units. Therefore, the value of $\sin x^\circ$ is $\frac{16}{23}$. Note that 16/23, .6956, .6957, 0.695, and 0.696 are examples of ways to enter a correct answer.

Question ID: 69b0d79d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Point O is the center of the circle above, and the measure of $\angle OAB$ is 30° . If the length of $\overline{OC}$ is 18, what is the length of arc $\widehat{AB}$?

Answer

- A. 9π
- B. 12π
- C. 15π
- D. 18π

Correct Answer: B

Rationale

Choice B is correct. Because segments OA and OB are radii of the circle centered at point O, these segments have equal lengths. Therefore, triangle AOB is an isosceles triangle, where angles OAB and OBA are congruent base angles of the triangle. It's given that angle OAB measures 30° . Therefore, angle OBA also measures 30° . Let x° represent the measure of angle AOB. Since the sum of the measures of the three angles of any triangle is 180° , it follows that $30^\circ + 30^\circ + x^\circ = 180^\circ$, or $60^\circ + x^\circ = 180^\circ$. Subtracting 60° from both sides of this equation yields $x^\circ = 120^\circ$, or $\frac{2\pi}{3}$ radians. Therefore, the measure of angle AOB, and thus the measure of arc $\widehat{AB}$, is $\frac{2\pi}{3}$ radians. Since $\overline{OC}$ is a radius of the given circle and its length is 18, the length of the radius of the circle is 18. Therefore, the length of arc $\widehat{AB}$ can be calculated as $\left(\frac{2\pi}{3}\right)(18)$, or 12π .

Choices A, C, and D are incorrect and may result from conceptual or computational errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is **648** square inches, and the area of rectangle $EFGH$ is **72** square inches. The length of the longest side of rectangle $ABCD$ is **36** inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

Answer

- A. **4**
- B. **9**
- C. **12**
- D. **36**

Correct Answer: C

Rationale

Choice C is correct. It's given that rectangle $ABCD$ is similar to rectangle $EFGH$. Therefore, if the length of each side of rectangle $ABCD$ is k times the length of the corresponding side of rectangle $EFGH$, then the area of rectangle $ABCD$ is k^2 times the area of rectangle $EFGH$. It's given that the area of rectangle $ABCD$ is **648** square inches and the area of rectangle $EFGH$ is **72** square inches. It follows that $k^2 = \frac{648}{72}$, or $k^2 = 9$. Taking the square root of each side of this equation yields $k = \sqrt{9}$, or $k = 3$. It follows that the length of each side of rectangle $ABCD$ is **3** times the length of the corresponding side of rectangle $EFGH$. It's given that the length of the longest side of rectangle $ABCD$ is **36** inches. Therefore, **36** inches is **3** times the length of the longest side of rectangle $EFGH$, and the longest side of rectangle $EFGH$ is equal to $\frac{36}{3}$, or **12**, inches.

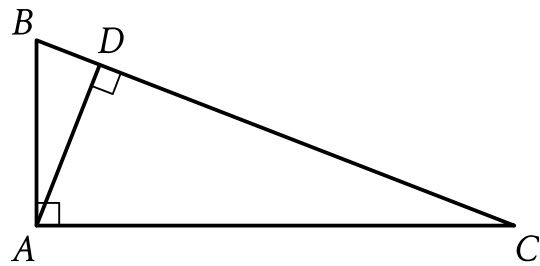
Choice A is incorrect. This is the length, in inches, of the longest side of a rectangle with side lengths that are $\frac{1}{9}$ the corresponding side lengths of rectangle $ABCD$, rather than a rectangle with an area that is $\frac{1}{9}$ the area of rectangle $ABCD$.

Choice B is incorrect. This is the factor by which the area of rectangle $ABCD$ is larger than the area of rectangle $EFGH$, not the length, in inches, of the longest side of rectangle $EFGH$.

Choice D is incorrect. This is the length, in inches, of the longest side of rectangle $ABCD$, not rectangle $EFGH$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{AD}$ and $\overline{BC}$ intersect at point D . If the length of $\overline{BD}$ is 5 and the tangent of angle CAD is 1.3, what is the length of $\overline{AD}$?

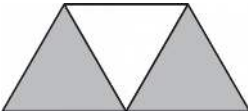
Correct Answer: 6.5, 13/2

Rationale

The correct answer is 6.5. The figure shows that $\triangle BAC$ is a right triangle. In the figure, $\overline{AD}$ is perpendicular to $\overline{BC}$, which forms two additional right triangles: $\triangle BDA$ and $\triangle ADC$. If two pairs of corresponding angles in two triangles are congruent, then the triangles are similar. $\triangle BDA$ shares angle B with $\triangle BAC$ and $\triangle ADC$ shares angle C with $\triangle BAC$. Therefore, the figure contains three similar right triangles: $\triangle BAC$, $\triangle BDA$, and $\triangle ADC$. In similar triangles, the tangents of corresponding angles are equivalent. It's given that the tangent of angle CAD is 1.3. Since angle CAD in $\triangle ADC$ corresponds to angle ABD in $\triangle BDA$, it follows that the tangent of angle ABD is also 1.3. The tangent of an acute angle in a right triangle is equal to the length of the leg opposite the angle divided by the length of the leg adjacent to the angle. Therefore, $\tan ABD = \frac{AD}{BD}$. It's given that the length of $\overline{BD}$ is 5. Substituting 5 for BD and 1.3 for $\tan ABD$ in the equation $\tan ABD = \frac{AD}{BD}$ yields $1.3 = \frac{AD}{5}$. Multiplying both sides of this equation by 5 yields $6.5 = AD$. Therefore, the length of $\overline{AD}$ is 6.5. Note that 6.5 and 13/2 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

Answer

- A. $2\sqrt{3}$
- B. $4\sqrt{3}$
- C. $8\sqrt{3}$
- D. 16

Correct Answer: C

Rationale

Choice C is correct. It's given that the logo is in the shape of a trapezoid that consists of three congruent equilateral triangles, and that the perimeter of the trapezoid is 20 centimeters (cm). Since the perimeter of the trapezoid is the sum of the lengths of 5 of the sides of the triangles, the length of each side of an equilateral triangle is $\frac{20}{5} = 4$ cm. Dividing up one equilateral triangle into two right triangles yields a pair of congruent 30°-60°-90° triangles. The shorter leg of each right triangle is half the length of the side of an equilateral triangle, or 2 cm. Using the Pythagorean Theorem, $a^2 + b^2 = c^2$, the height of the equilateral triangle can be found. Substituting $a = 2$ and $c = 4$ and solving for b yields $\sqrt{4^2 - 2^2} = \sqrt{12} = 2\sqrt{3}$ cm. The area of one equilateral triangle is $\frac{1}{2}bh$, where $b = 2$ and $h = 2\sqrt{3}$. Therefore, the area of one equilateral triangle is $\frac{1}{2}(4)(2\sqrt{3}) = 4\sqrt{3}$ cm². The shaded area consists of two such triangles, so its area is $(2)(4)\sqrt{3} = 8\sqrt{3}$ cm².

Alternate approach: The area of a trapezoid can be found by evaluating the expression $\frac{1}{2}(b_1 + b_2)h$, where b_1 is the length of one base, b_2 is the length of the other base, and h is the height of the trapezoid. Substituting $b_1 = 8$, $b_2 = 4$, and $h = 2\sqrt{3}$ yields the expression $\frac{1}{2}(8 + 4)(2\sqrt{3})$, or $\frac{1}{2}(12)(2\sqrt{3})$, which gives an area of $12\sqrt{3}$ cm² for the trapezoid. Since two-thirds of the trapezoid is shaded, the area of the shaded region is $\frac{2}{3} \times 12\sqrt{3} = 8\sqrt{3}$.

Choice A is incorrect. This is the height of the trapezoid. Choice B is incorrect. This is the area of one of the equilateral triangles, not two. Choice D is incorrect and may result from using a height of 4 for each triangle rather than the height of $2\sqrt{3}$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Circle A in the xy -plane has the equation $(x + 5)^2 + (y - 5)^2 = 4$. Circle B has the same center as circle A. The radius of circle B is two times the radius of circle A. The equation defining circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = k$, where k is a constant. What is the value of k ?

Correct Answer: 16

Rationale

The correct answer is **16**. An equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - u)^2 = r^2$, where the center of the circle is (t, u) , the radius of the circle is r , and where t , u , and r are constants. It's given that the equation of circle A is $(x + 5)^2 + (y - 5)^2 = 4$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 2^2$. Therefore, the center of circle A is $(-5, 5)$ and the radius of circle A is **2**. It's given that circle B has the same center as circle A and that the radius of circle B is two times the radius of circle A. Therefore, the center of circle B is $(-5, 5)$ and the radius of circle B is **2(2)**, or **4**. Substituting -5 for t , 5 for u , and **4** for r into the equation $(x - t)^2 + (y - u)^2 = r^2$ yields $(x + 5)^2 + (y - 5)^2 = 4^2$, which is equivalent to $(x + 5)^2 + (y - 5)^2 = 16$. It follows that the equation of circle B in the xy -plane is $(x + 5)^2 + (y - 5)^2 = 16$. Therefore, the value of k is **16**.

Question ID: f389569d

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

Answer

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

Correct Answer: C

Rationale

Choice C is correct. It’s given that the equilateral triangle has a height of $39\sqrt{3}$. In equilateral triangles, all three angles have a measure of 60° and the height is equal to the length of an altitude from one vertex to the midpoint of the side across from the vertex. Thus, an altitude in this equilateral triangle has a length of $39\sqrt{3}$. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the 60° angle in each 30-60-90 right triangle. Since the altitude has a length of $39\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each 30° angle has a length of 39 and each hypotenuse has a length of $2(39)$, or 78. Therefore, each side of the equilateral triangle has a length of 78. In equilateral triangles, all three sides have equal length. Therefore, the perimeter of this equilateral triangle is $78 + 78 + 78$, or 234.

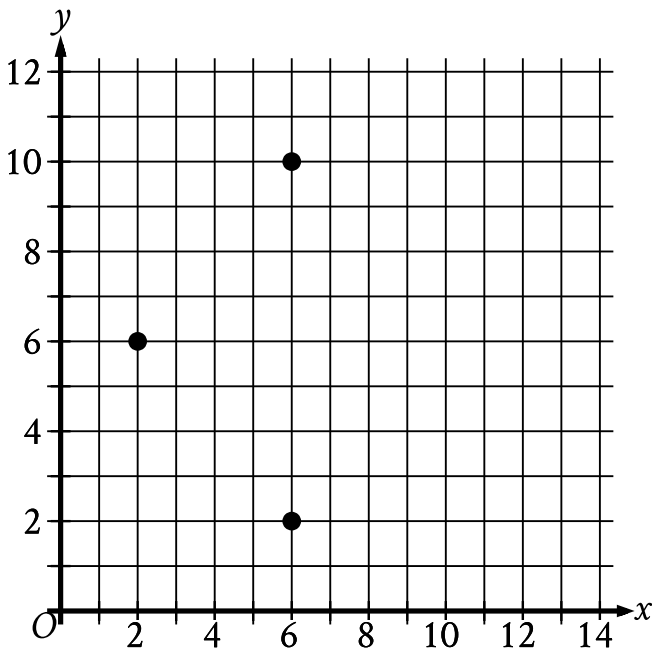
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

Correct Answer: 8

Rationale

The correct answer is 8. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point (6, 6) is halfway between the points (6, 2) and (6, 10), and is a distance of 4 units from each of those two points. The point (6, 6) is also a distance of 4 units from (2, 6). Because the point (6, 6) is the same distance from all three points shown, it must be the center of the circle. Since that distance is 4, it follows that the radius of the circle is 4. The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the given circle is $2\pi(4)$, or 8π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is 8.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Acute angle P has a measure of p radians, and acute angle T has a measure of t radians. The function g is defined by $g(x) = p(8)^x + 6t$. If $\sin p = \cos t$, which of the following represents the value $g(0)$ in terms of t ?

Answer

- A. $6t + \pi$
- B. $6t$
- C. $5t + \pi$
- D. $5t + \frac{\pi}{2}$

Correct Answer: D

Rationale

Choice D is correct. It's given that acute angle P has a measure of p radians and that acute angle T has a measure of t radians. If $\sin p = \cos t$, it follows that $p + t = \frac{\pi}{2}$, or $p = \frac{\pi}{2} - t$. It's also given that function g is defined by $g(x) = p(8)^x + 6t$. To determine the value of $g(0)$ in terms of t , substituting 0 for x and $\frac{\pi}{2} - t$ for p in the equation $g(x) = p(8)^x + 6t$ yields $g(0) = (\frac{\pi}{2} - t)(8)^0 + 6t$, or $g(0) = (\frac{\pi}{2} - t)(1) + 6t$. This is equivalent to $g(0) = \frac{\pi}{2} - t + 6t$, or $g(0) = 5t + \frac{\pi}{2}$. Therefore, $5t + \frac{\pi}{2}$ represents the value of $g(0)$ in terms of t .

Choice A is incorrect and may result from conceptual or calculation errors.

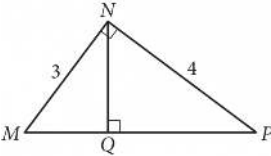
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 740bf79f

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, what is the length of $\overline{NQ}$?

Answer

- A. 2.2
- B. 2.3
- C. 2.4
- D. 2.5

Correct Answer: C

Rationale

Choice C is correct. First, $\overline{MP}$ is the hypotenuse of right $\triangle MNP$, whose legs have lengths 3 and 4. Therefore, $(MP)^2 = 3^2 + 4^2$, so $(MP)^2 = 25$ and $MP = 5$. Second, because $\angle MNP$ corresponds to $\angle NQP$ and because $\angle MPN$ corresponds to $\angle NPQ$, $\triangle MNP$ is similar to $\triangle NQP$. The ratio of corresponding sides of similar triangles is constant, so $\frac{NQ}{MN} = \frac{NP}{MP}$. Since $MP = 5$ and it's given that $MN = 3$ and $NP = 4$, $\frac{NQ}{3} = \frac{4}{5}$. Solving for NQ results in $NQ = \frac{12}{5}$, or 2.4.

Choices A, B, and D are incorrect and may result from setting up incorrect ratios.

Question ID: ab176ad6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The equation $(x + 6)^2 + (y + 3)^2 = 121$ defines a circle in the xy-plane. What is the radius of the circle?

Rationale

The correct answer is 11. A circle with equation $(x - a)^2 + (y - b)^2 = r^2$, where a, b, and r are constants, has center (a,b) and radius r.

Therefore, the radius of the given circle is $\sqrt{121}$, or 11.

Question ID: 179edb12

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a height of **22 centimeters (cm)** and a base with a diameter of **18 cm**. The volume of this cone is $n\pi \text{ cm}^3$. What is the value of n ?

Correct Answer: 594

Rationale

The correct answer is **594**. It's given that the right circular cone has a height of **22 cm** and a base with a diameter of **18 cm**. Since the diameter of the base of the cone is **18 cm**, the radius of the base is **9 cm**. The volume V , in cm^3 , of a right circular cone can be found using the formula $V = \frac{1}{3}\pi r^2 h$, where h is the height, in **cm**, of the cone and r is the radius, in **cm**, of the base of the cone. Substituting **22** for h and **9** for r in this formula yields $V = \frac{1}{3}\pi(9)^2(22)$, or $V = 594\pi$. Therefore, the volume of the cone is **$594\pi \text{ cm}^3$** . It's given that the volume of the cone is $n\pi \text{ cm}^3$. Therefore, the value of n is **594**.

Question ID: 3e577e4a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has its center at $(-4, -6)$. Line k is tangent to this circle at the point $(-7, -7)$. What is the slope of line k ?

Answer

- A. -3
- B. $-\frac{1}{3}$
- C. $\frac{1}{3}$
- D. 3

Correct Answer: A

Rationale

Choice A is correct. A line that's tangent to a circle is perpendicular to the radius of the circle at the point of tangency. It's given that the circle has its center at $(-4, -6)$ and line k is tangent to the circle at the point $(-7, -7)$. The slope of a radius defined by the points (q, r) and (s, t) can be calculated as $\frac{t-r}{s-q}$. The points $(-7, -7)$ and $(-4, -6)$ define the radius of the circle at the point of tangency. Therefore, the slope of this radius can be calculated as $\frac{(-6)-(-7)}{(-4)-(-7)}$, or $\frac{1}{3}$. If a line and a radius are perpendicular, the slope of the line must be the negative reciprocal of the slope of the radius. The negative reciprocal of $\frac{1}{3}$ is -3 . Thus, the slope of line k is -3 .

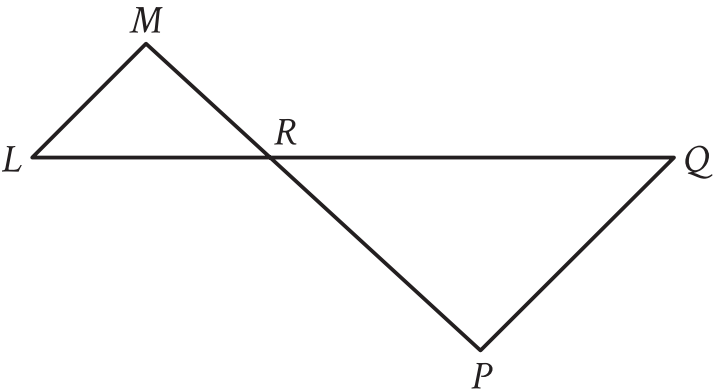
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the slope of the radius of the circle at the point of tangency, not the slope of line k .

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

Answer

- A. $\frac{576}{49}$
- B. $\frac{144}{7}$
- C. 63
- D. $\frac{441}{4}$

Correct Answer: D

Rationale

Choice D is correct. The figure shows that $\angle MRL$ and $\angle PRQ$ are vertical angles. Since vertical angles are congruent, $\angle MRL$ and $\angle PRQ$ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, $\angle MLR$ and $\angle PQR$ are congruent. Since $\triangle LMR$ and $\triangle PQR$ have two pairs of congruent angles, the triangles are similar, where $\overline{MR}$ in $\triangle LMR$ corresponds to $\overline{PR}$ in $\triangle PQR$. Since the areas of similar triangles are proportional to the squares of the lengths of corresponding sides, it follows that $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$. It's given that the lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively, and the area of $\triangle LMR$ is 36 square units. Substituting 14 for RP , 8 for MR , and 36 for $\text{area of } \triangle LMR$ in the equation $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$ yields $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{14}{8}\right)^2$, which is equivalent to $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{7}{4}\right)^2$, or $\frac{\text{area of } \triangle PQR}{36} = \frac{49}{16}$. Multiplying each side of this equation by 36 yields $\text{area of } \triangle PQR = 36\left(\frac{49}{16}\right)$, or $\text{area of } \triangle PQR = \frac{441}{4}$. Therefore, the area of $\triangle PQR$ is $\frac{441}{4}$ square units.

Choice A is incorrect. This is the result of multiplying 36 by $\left(\frac{8}{14}\right)^2$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice B is incorrect. This is the result of multiplying 36 by $\frac{8}{14}$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice C is incorrect. This is the result of multiplying **36** by $\frac{14}{8}$ instead of by $\left(\frac{14}{8}\right)^2$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V , in cubic inches, of such containers, where r is the radius, in inches?

Answer

- A. $V = 4\pi^3$
- B. $V = \pi(2r)^3$
- C. $V = \pi^2 + 4\pi$
- D. $V = \pi^3 + 4\pi^2$

Correct Answer: D

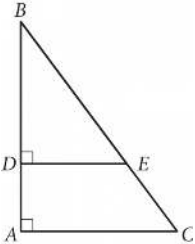
Rationale

Choice D is correct. The volume, V , of a right cylinder is given by the formula $V = \pi r^2 h$, where r represents the radius of the base of the cylinder and h represents the height. Since the height is 4 inches longer than the radius, the expression $r + 4$ represents the height of each cylindrical container. It follows that the volume of each container is represented by the equation $V = \pi r^2 (r + 4)$. Distributing the expression πr^2 into each term in the parentheses yields $V = \pi r^3 + 4\pi r^2$.

Choice A is incorrect and may result from representing the height as $4r$ instead of $r + 4$. Choice B is incorrect and may result from representing the height as $2r$ instead of $r + 4$. Choice C is incorrect and may result from representing the volume of a right cylinder as $V = \pi r h$ instead of $V = \pi r^2 h$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

Rationale

The correct answer is 6. Since $\tan B = \frac{3}{4}$, $\triangle ABC$ and $\triangle DBE$ are both similar to 3-4-5 triangles. This means that they are both similar to the right triangle with sides of lengths 3, 4, and 5. Since $BC = 15$, which is 3 times as long as the hypotenuse of the 3-4-5 triangle, the similarity ratio of $\triangle ABC$ to the 3-4-5 triangle is 3:1. Therefore, the length of $\overline{AC}$ (the side opposite to $\angle B$) is $3 \times 3 = 9$, and the length of $\overline{AB}$ (the side adjacent to $\angle B$) is $4 \times 3 = 12$. It is also given that $DA = 4$. Since $AB = DA + DB$ and $AB = 12$, it follows that $DB = 8$, which means that the similarity ratio of $\triangle DBE$ to the 3-4-5 triangle is 2:1 ($\overline{DB}$ is the side adjacent to $\angle B$). Therefore, the length of $\overline{DE}$, which is the side opposite to $\angle B$, is $3 \times 2 = 6$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

Answer

- A. $-14x + 1,540$
- B. $14x - 320$
- C. $-28x + 1,720$
- D. 360

Correct Answer: A

Rationale

Choice A is correct. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. Since there are two parallel lines intersected by a transversal, all four acute angles have the same measure and all four obtuse angles have the same measure. Additionally, each acute angle is supplementary to each obtuse angle. It's given that the measure of one of these eight angles is $(7x - 250)^\circ$. It follows that a supplementary angle has measure $(180 - (7x - 250))^\circ$, or $(-7x + 430)^\circ$. It's also given that the sum of the measures of four of the eight angles is k° . It follows that the possible values of k are $4(7x - 250)$; $(7x - 250) + 3(-7x + 430)$; $2(7x - 250) + 2(-7x + 430)$; $3(7x - 250) + (-7x + 430)$; and $4(-7x + 430)$. These values are equivalent to $28x - 1,000$; $-14x + 1,040$; 360 ; $14x - 320$; and $-28x + 1,720$, respectively. It follows that of the given choices, only $-14x + 1,540$ could NOT be equivalent to k , for all values of x .

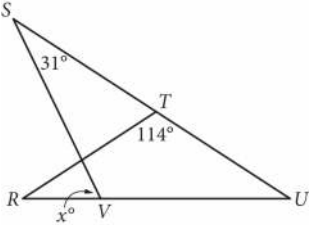
Choice B is incorrect. This is the sum of three angles with measure $(7x - 250)^\circ$ and one angle with measure $(-7x + 430)^\circ$.

Choice C is incorrect. This is the sum of four angles with measure $(-7x + 430)^\circ$.

Choice D is incorrect. This is the sum of two angles with measure $(7x - 250)^\circ$ and two angles with measure $(-7x + 430)^\circ$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, $RT = TU$. What is the value of x ?

Answer

- A. 72
- B. 66
- C. 64
- D. 58

Correct Answer: C

Rationale

Choice C is correct. Since $RT = TU$, it follows that $\triangle RTU$ is an isosceles triangle with base RU. Therefore, $\angle TRU$ and $\angle TUR$ are the base angles of an isosceles triangle and are congruent. Let the measures of both $\angle TRU$ and $\angle TUR$ be t° . According to the triangle sum theorem, the sum of the measures of the three angles of a triangle is 180° . Therefore, $114^\circ + 2t^\circ = 180^\circ$, so $t = 33$.

Note that $\angle TUR$ is the same angle as $\angle SUV$. Thus, the measure of $\angle SUV$ is 33° . According to the triangle exterior angle theorem, an external angle of a triangle is equal to the sum of the opposite interior angles. Therefore, x° is equal to the sum of the measures of $\angle VSU$ and $\angle SUV$; that is, $31^\circ + 33^\circ = 64^\circ$. Thus, the value of x is 64.

Choice B is incorrect. This is the measure of $\angle STR$, but $\angle STR$ is not congruent to $\angle SVR$. Choices A and D are incorrect and may result from a calculation error.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is **2** times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is **5** square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

Answer

- A. **7**
- B. **10**
- C. **20**
- D. **25**

Correct Answer: C

Rationale

Choice C is correct. It’s given that parallelogram $ABCD$ is similar to parallelogram $PQRS$. When two parallelograms are similar, if the scale factor between their corresponding side lengths is k , the scale factor between their areas is k^2 . It’s given that the length of each side of parallelogram $PQRS$ is **2** times the length of its corresponding side of parallelogram $ABCD$, so the scale factor between their corresponding side lengths is **2**. It follows that the scale factor between their areas is **2²**, or **4**. It’s given that the area, in square centimeters, of parallelogram $ABCD$ is **5**. It follows that the area, in square centimeters, of parallelogram $PQRS$ is **5(4)**, or **20**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 9e44284b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, the graph of $2x^2 - 6x + 2y^2 + 2y = 45$ is a circle. What is the radius of the circle?

Answer

- A. 5
- B. 6.5
- C. $\sqrt{40}$
- D. $\sqrt{50}$

Correct Answer: A

Rationale

Choice A is correct. One way to find the radius of the circle is to rewrite the given equation in standard form, $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and the radius of the circle is r . To do this, divide the original equation, $2x^2 - 6x + 2y^2 + 2y = 45$, by 2 to make the leading coefficients of x^2 and y^2 each equal to 1: $x^2 - 3x + y^2 + y = 22.5$. Then complete the square to put the equation in standard form. To do so, first rewrite $x^2 - 3x + y^2 + y = 22.5$ as $(x^2 - 3x + 2.25) - 2.25 + (y^2 + y + 0.25) - 0.25 = 22.5$. Second, add 2.25 and 0.25 to both sides of the equation: $(x^2 - 3x + 2.25) + (y^2 + y + 0.25) = 25$. Since $x^2 - 3x + 2.25 = (x - 1.5)^2$, $y^2 + y + 0.25 = (y + 0.5)^2$, and $25 = 5^2$, it follows that $(x - 1.5)^2 + (y + 0.5)^2 = 5^2$. Therefore, the radius of the circle is 5.

Choices B, C, and D are incorrect and may be the result of errors in manipulating the equation or of a misconception about the standard form of the equation of a circle in the xy -plane.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle XYZ , angles X and Z are acute angles. Point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. The measure of $\angle XZY$ is a° , the measure of $\angle XYR$ is b° , and the length of $\overline{XY}$ is 19. If $\sin a^\circ = \frac{12}{13}$, what is the value of $\tan b^\circ$?

Correct Answer: 2.4, 12/5

Rationale

The correct answer is $\frac{12}{5}$. It's given that in right triangle XYZ , $\angle X$ and $\angle Z$ are acute angles. It follows that $\angle Y$ is a right angle. It's given that point R lies on $\overline{XZ}$ such that $\overline{YR}$ is perpendicular to $\overline{XZ}$. Since $\angle Y$ is a right angle and is comprised of $\angle XYR$ and $\angle ZYR$, it follows that the sum of the measures of $\angle XYR$ and $\angle ZYR$ is 90° . It's given that the measure of $\angle XYR$ is b° . It follows that the measure of $\angle ZYR$ is $(90 - b)^\circ$. Since $\overline{YR}$ is perpendicular to $\overline{XZ}$, it follows that $\angle XRY$ and $\angle ZRY$ are right angles. It follows that $\triangle XRY$ and $\triangle ZRY$ are right triangles. It's given that the measure of $\angle XZY$ is a° . Because the sum of the measures of the interior angles of a triangle is 180° , it follows that the measures of $\angle ZRY$, $\angle XZY$, and $\angle ZYR$ have a sum of 180° . Since the measure of $\angle ZRY$ is 90° , the measure of $\angle YZR$ is a° , and the measure of $\angle ZYR$ is $(90 - b)^\circ$, it follows that $90 + a + 90 - b = 180$, or $a - b + 180 = 180$. Subtracting 180 from both sides of this equation yields $a - b = 0$. Adding b to both sides of this equation yields $a = b$. It's given that $\sin a^\circ = \frac{12}{13}$. Since $a = b$, it follows that $\sin b^\circ = \frac{12}{13}$. In a right triangle, the sine of an acute angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse. It follows that, for a positive constant k , $12k$ represents the length of the leg opposite $\angle XYR$ and $13k$ represents the length of the hypotenuse of $\triangle XRY$. The Pythagorean theorem states that in a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the other two sides. Let nk represent the length of the leg adjacent to $\angle XYR$. It follows that $(13k)^2 = (12k)^2 + (nk)^2$, or $169k^2 = 144k^2 + n^2k^2$. Subtracting $144k^2$ from both sides of this equation yields $25k^2 = n^2k^2$. Taking the positive square root of both sides of this equation yields $5k = nk$. Therefore, the length of the leg adjacent to $\angle XYR$ is $5k$. In a right triangle, the tangent of an acute angle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan b^\circ = \frac{12k}{5k}$, or $\tan b^\circ = \frac{12}{5}$. Note that 12/5 and 2.4 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

Answer

- A. $\frac{9}{7,744}$
- B. $\frac{27}{3,872}$
- C. $\frac{3}{88}$
- D. $\frac{9}{44}$

Correct Answer: A

Rationale

Choice A is correct. Since all cubes are similar to each other, cube X and cube Y are similar figures. In similar figures, if lengths differ by a scale factor of k , then areas differ by a scale factor of k^2 . It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. It follows that the surface area, in square inches, of cube Y is $\left(\frac{3}{88}\right)^2$, or $\frac{9}{7,744}$, times the surface area, in square inches, of cube X. Therefore, the value of n is $\frac{9}{7,744}$.

Alternate approach: The surface area of a cube is $6s^2$, where s is the edge length. Let x represent the edge length, in inches, of cube X. Substituting x for s in $6s^2$ yields $6x^2$. Therefore, the surface area, in square inches, of cube X can be expressed as $6x^2$. It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. Therefore, the edge length, in inches, of cube Y can be expressed as $\left(\frac{3}{88}\right)x$, or $\frac{3x}{88}$. Substituting $\frac{3x}{88}$ for s in $6s^2$ yields $6\left(\frac{3x}{88}\right)^2$, which is equivalent to $6\left(\frac{9x^2}{7,744}\right)$, or $\frac{27x^2}{3,872}$. Therefore, the surface area, in square inches, of cube Y can be expressed as $\frac{27x^2}{3,872}$. It's given that the surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. It follows that $\frac{27x^2}{3,872} = n(6x^2)$. Dividing both sides of this equation by $6x^2$ yields $\frac{27x^2}{3,872(6x^2)} = n$, which is equivalent to $\frac{27x^2}{23,232x^2} = n$, or $\frac{9}{7,744} = n$. Therefore, the value of n is $\frac{9}{7,744}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

An isosceles right triangle has a perimeter of $94 + 94\sqrt{2}$ inches. What is the length, in inches, of one leg of this triangle?

Answer

- A. 47
- B. $47\sqrt{2}$
- C. 94
- D. $94\sqrt{2}$

Correct Answer: B

Rationale

Choice B is correct. It's given that the right triangle is isosceles. In an isosceles right triangle, the two legs have equal lengths, and the length of the hypotenuse is $\sqrt{2}$ times the length of one of the legs. Let ℓ represent the length, in inches, of each leg of the isosceles right triangle. It follows that the length of the hypotenuse is $\ell\sqrt{2}$ inches. The perimeter of a figure is the sum of the lengths of the sides of the figure. Therefore, the perimeter of the isosceles right triangle is $\ell + \ell + \ell\sqrt{2}$ inches. It's given that the perimeter of the triangle is $94 + 94\sqrt{2}$ inches. It follows that $\ell + \ell + \ell\sqrt{2} = 94 + 94\sqrt{2}$. Factoring the left-hand side of this equation yields $(1 + 1 + \sqrt{2})\ell = 94 + 94\sqrt{2}$, or $(2 + \sqrt{2})\ell = 94 + 94\sqrt{2}$. Dividing both sides of this equation by $2 + \sqrt{2}$ yields $\ell = \frac{94+94\sqrt{2}}{2+\sqrt{2}}$. Rationalizing the denominator of the right-hand side of this equation by multiplying the right-hand side of the equation by $\frac{2-\sqrt{2}}{2-\sqrt{2}}$ yields $\ell = \frac{(94+94\sqrt{2})(2-\sqrt{2})}{(2+\sqrt{2})(2-\sqrt{2})}$. Applying the distributive property to the numerator and to the denominator of the right-hand side of this equation yields $\ell = \frac{188-94\sqrt{2}+188\sqrt{2}-94\sqrt{4}}{4-2\sqrt{2}+2\sqrt{2}-\sqrt{4}}$. This is equivalent to $\ell = \frac{94\sqrt{2}}{2}$, or $\ell = 47\sqrt{2}$. Therefore, the length, in inches, of one leg of the isosceles right triangle is $47\sqrt{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length, in inches, of the hypotenuse.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Quadrilaterals $PQRS$ and $WXYZ$ are similar, where P , Q , and R correspond to W , X , and Y , respectively. The measure of $\angle S$ is 135° , $PS = 45$, and $WZ = 9$. What is the measure of $\angle Z$?

Answer

- A. 5°
- B. 27°
- C. 45°
- D. 135°

Correct Answer: D

Rationale

Choice D is correct. Corresponding angles in similar figures have equal measure. It's given that quadrilaterals $PQRS$ and $WXYZ$ are similar and that P , Q , and R correspond to W , X , and Y . It follows that $\angle S$ corresponds to $\angle Z$. It's also given that the measure of $\angle S$ is 135° . Therefore, the measure of $\angle Z$ is 135° .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This is the supplement of the measure of $\angle Z$, not the measure of $\angle Z$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

$RS = 440$

$ST = 384$

$TR = 584$

The side lengths of right triangle RST are given. Triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . What is the value of $\tan W$?

Answer

- A. $\frac{48}{73}$
- B. $\frac{55}{73}$
- C. $\frac{48}{55}$
- D. $\frac{55}{48}$

Correct Answer: D

Rationale

Choice D is correct. The hypotenuse of triangle RST is the longest side and is across from the right angle. The longest side length given is **584**, which is the length of side TR . Therefore, the hypotenuse of triangle RST is side TR , so the right angle is angle S . The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side, which is the side across from the angle, to the length of the adjacent side, which is the side closest to the angle that is not the hypotenuse. It follows that the opposite side of angle T is side RS and the adjacent side of angle T is side ST . Therefore, $\tan T = \frac{RS}{ST}$. Substituting **440** for RS and **384** for ST in this equation yields $\tan T = \frac{440}{384}$. This is equivalent to $\tan T = \frac{55}{48}$. It's given that triangle RST is similar to triangle UVW , where S corresponds to V and T corresponds to W . It follows that R corresponds to U . Therefore, the hypotenuse of triangle UVW is side WU , which means $\tan W = \frac{UV}{VW}$. Since the lengths of corresponding sides of similar triangles are proportional, $\frac{RS}{ST} = \frac{UV}{VW}$. Therefore, $\tan W = \frac{UV}{VW}$ is equivalent to $\tan W = \frac{RS}{ST}$, or $\tan W = \tan T$. Thus, $\tan W = \frac{55}{48}$.

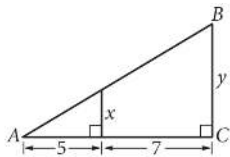
Choice A is incorrect. This is the value of $\cos W$, not $\tan W$.

Choice B is incorrect. This is the value of $\sin W$, not $\tan W$.

Choice C is incorrect. This is the value of $\frac{1}{\tan W}$, not $\tan W$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

Rationale

The correct answer is either $\frac{10}{3}$, $\frac{15}{4}$, or $\frac{25}{6}$. The area of triangle ABC can be expressed as $\frac{1}{2}(5+7)y$ or $6y$. It's given that the area of triangle ABC is at least 48 but no more than 60. It follows that $48 \leq 6y \leq 60$. Dividing by 6 to isolate y in this compound inequality yields $8 \leq y \leq 10$. Since y is an integer, $y = 8, 9, \text{ or } 10$. In the given figure, the two right triangles shown are similar because they have two pairs of congruent angles: their respective right angles and angle A. Therefore, the following proportion is true: $\frac{x}{y} = \frac{5}{12}$. Substituting 8 for y in the proportion results in $\frac{x}{8} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{10}{3}$. Substituting 9 for y in the proportion results in $\frac{x}{9} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{15}{4}$. Substituting 10 for y in the proportion results in $\frac{x}{10} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{25}{6}$. Note that 10/3, 15/4, 25/6, 3.333, 3.75, 4.166, and 4.167 are examples of ways to enter a correct answer.

Question ID: 5b2b8866

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A rectangular poster has an area of **360** square inches. A copy of the poster is made in which the length and width of the original poster are each increased by **20%**. What is the area of the copy, in square inches?

Correct Answer: 2592/5, 518.4

Rationale

The correct answer is **518.4**. It's given that the area of the original poster is **360** square inches. Let ℓ represent the length, in inches, of the original poster, and let w represent the width, in inches, of the original poster. Since the area of a rectangle is equal to its length times its width, it follows that **360** = ℓw . It's also given that a copy of the poster is made in which the length and width of the original poster are each increased by **20%**. It follows that the length of the copy is the length of the original poster plus **20%** of the length of the original poster, which is equivalent to $\ell + \frac{20}{100}\ell$ inches. This length can be rewritten as $\ell + 0.2\ell$ inches, or **1.2** ℓ inches. Similarly, the width of the copy is the width of the original poster plus **20%** of the width of the original poster, which is equivalent to $w + \frac{20}{100}w$ inches. This width can be rewritten as $w + 0.2w$ inches, or **1.2** w inches. Since the area of a rectangle is equal to its length times its width, it follows that the area, in square inches, of the copy is equal to **(1.2** ℓ **)(1.2** w **)**, which can be rewritten as **(1.2)(1.2)(** ℓw **)**. Since **360** = ℓw , the area, in square inches, of the copy can be found by substituting **360** for ℓw in the expression **(1.2)(1.2)(** ℓw **)**, which yields **(1.2)(1.2)(360)**, or **518.4**. Therefore, the area of the copy, in square inches, is **518.4**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has its center at $(16, 17)$ and has a radius of $7k$. Which equation represents this circle?

Answer

- A. $(x - 16)^2 + (y - 17)^2 = 49k$
- B. $(x - 16)^2 + (y - 17)^2 = 49k^2$
- C. $(x - 16)^2 + (y - 17)^2 = 7k$
- D. $(x - 16)^2 + (y - 17)^2 = 7k^2$

Correct Answer: B

Rationale

Choice B is correct. The equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that this circle has a center at $(16, 17)$ and a radius of $7k$. Substituting 16 for h , 17 for k , and $7k$ for r in $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 16)^2 + (y - 17)^2 = (7k)^2$, or $(x - 16)^2 + (y - 17)^2 = 49k^2$. Therefore, the equation that represents this circle is $(x - 16)^2 + (y - 17)^2 = 49k^2$.

Choice A is incorrect. This equation represents a circle with radius $7\sqrt{k}$, not $7k$.

Choice C is incorrect. This equation represents a circle with radius $\sqrt{7k}$, not $7k$.

Choice D is incorrect. This equation represents a circle with radius $\sqrt{7k}$, not $7k$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a volume of $\frac{1}{3} \pi$ cubic feet and a height of 9 feet. What is the radius, in feet, of the base of the cone?

Answer

- A. $\frac{1}{3}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. 3

Correct Answer: A

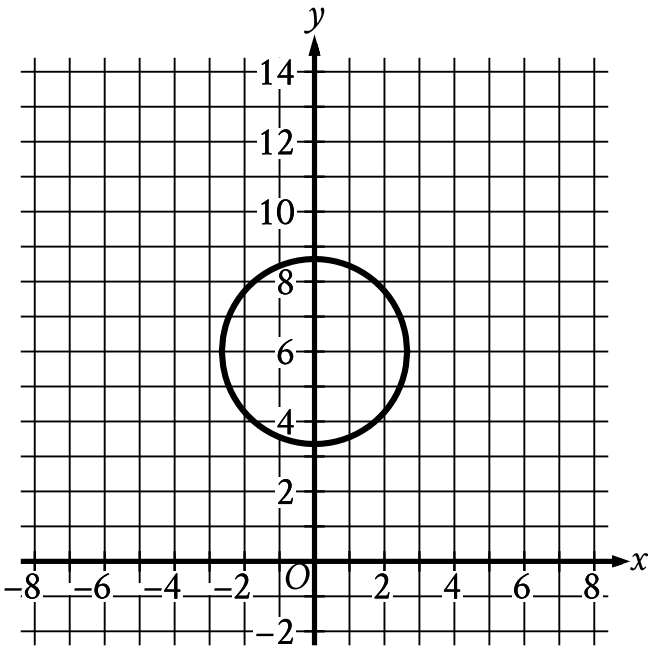
Rationale

Choice A is correct. The equation for the volume of a right circular cone is $V = \frac{1}{3} \pi r^2 h$. It's given that the volume of the right circular cone is $\frac{1}{3} \pi$ cubic feet and the height is 9 feet. Substituting these values for V and h, respectively, gives $\frac{1}{3} \pi = \frac{1}{3} \pi r^2 (9)$. Dividing both sides of the equation by $\frac{1}{3} \pi$ gives $1 = r^2 (9)$. Dividing both sides of the equation by 9 gives $\frac{1}{9} = r^2$. Taking the square root of both sides results in two possible values for the radius, $\sqrt{\left(\frac{1}{9}\right)}$ or $-\sqrt{\left(\frac{1}{9}\right)}$. Since the radius can't have a negative value, that leaves $\sqrt{\left(\frac{1}{9}\right)}$ as the only possibility. Applying the quotient property of square roots, $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$, results in $r = \frac{\sqrt{1}}{\sqrt{9}}$, or $r = \frac{1}{3}$.

Choices B and C are incorrect and may result from incorrectly evaluating $\sqrt{\left(\frac{1}{9}\right)}$. Choice D is incorrect and may result from solving $r^2 = 9$ instead of $r^2 = \frac{1}{9}$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question



Circle A shown is defined by the equation $x^2 + (y - 6)^2 = 7$. Circle B (not shown) has the same radius but is translated **96** units to the right. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, where h , k , and a are constants, what is the value of $4a$?

Correct Answer: 28

Rationale

The correct answer is **28**. The equation of a circle in the xy -plane can be written as $(x - t)^2 + (y - s)^2 = r^2$, where the center of the circle is (t, s) and the radius of the circle is r . It's given that circle A is defined by the equation $x^2 + (y - 6)^2 = 7$, which can be written as $(x - 0)^2 + (y - 6)^2 = (\sqrt{7})^2$. It follows that $r = \sqrt{7}$ and the radius of circle A is $\sqrt{7}$. It's also given that circle B has the same radius as circle A. If the equation of circle B is $(x - h)^2 + (y - k)^2 = a$, then $a = r^2$. Substituting $\sqrt{7}$ for r in this equation yields $a = (\sqrt{7})^2$, or $a = 7$. It follows that the value of $4a$ is $4(7)$, or **28**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

For triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH ?

- I. The measure of angle A is equal to the measure of angle F .
- II. The measure of angle C is equal to the measure of angle H .
- III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.

Answer

- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

Correct Answer: B

Rationale

Choice B is correct. It's given that in triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . For statement I, if the measure of angle A is equal to the measure of angle F , then triangles ABC and FGH have two pairs of congruent angles, angles A and F and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement I is sufficient. For statement II, if the measure of angle C is equal to the measure of angle H , then triangles ABC and FGH have 2 pairs of congruent angles, angles C and H and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement II is sufficient. For statement III, if the length of $\overline{FH}$ is 3 times the length of $\overline{AC}$, then the ratio of the length of $\overline{FH}$ to the length of $\overline{AC}$ is 3 . It's given that the length of $\overline{FG}$ is 18 and the length of $\overline{AB}$ is 6 . Therefore, the ratio of the length of $\overline{FG}$ to the length of $\overline{AB}$ is also 3 , so there are two pairs of sides in proportion. However, angle B isn't between sides AB and AC , and angle G isn't between sides FG and FH . Since the given angle measures aren't between the proportional sides, this information isn't sufficient to prove that the triangles are similar. Therefore, statements I and II are each sufficient, but statement III isn't sufficient.

Choice A is incorrect and may result from conceptual or calculation errors.

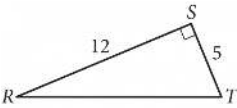
Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 6933b3d9

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In triangle RST above, point W (not shown) lies on $\overline{RT}$. What is the value of $\cos(\angle RSW) - \sin(\angle WST)$?

Rationale

The correct answer is 0. Note that no matter where point W is on $\overline{RT}$, the sum of the measures of $\angle RSW$ and $\angle WST$ is equal to the measure of $\angle RST$, which is 90° . Thus, $\angle RSW$ and $\angle WST$ are complementary angles. Since the cosine of an angle is equal to the sine of its complementary angle, $\cos(\angle RSW) = \sin(\angle WST)$. Therefore, $\cos(\angle RSW) - \sin(\angle WST) = 0$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Triangle ABC is similar to triangle DEF , where A corresponds to D and C corresponds to F . Angles C and F are right angles. If $\tan(A) = \sqrt{3}$ and $DF = 125$, what is the length of $\overline{DE}$?

Answer

- A. $125\frac{\sqrt{3}}{3}$
- B. $125\frac{\sqrt{3}}{2}$
- C. $125\sqrt{3}$
- D. 250

Correct Answer: D

Rationale

Choice D is correct. Corresponding angles in similar triangles have equal measures. It's given that triangle ABC is similar to triangle DEF , where A corresponds to D , so the measure of angle A is equal to the measure of angle D . Therefore, if $\tan(A) = \sqrt{3}$, then $\tan(D) = \sqrt{3}$. It's given that angles C and F are right angles, so triangles ABC and DEF are right triangles. The adjacent side of an acute angle in a right triangle is the side closest to the angle that is not the hypotenuse. It follows that the adjacent side of angle D is side DF . The opposite side of an acute angle in a right triangle is the side across from the acute angle. It follows that the opposite side of angle D is side EF . The tangent of an acute angle in a right triangle is the ratio of the length of the opposite side to the length of the adjacent side. Therefore, $\tan(D) = \frac{EF}{DF}$. If $DF = 125$, the length of side EF can be found by substituting $\sqrt{3}$ for $\tan(D)$ and 125 for DF in the equation $\tan(D) = \frac{EF}{DF}$, which yields $\sqrt{3} = \frac{EF}{125}$. Multiplying both sides of this equation by 125 yields $125\sqrt{3} = EF$. Since the length of side EF is $\sqrt{3}$ times the length of side DF , it follows that triangle DEF is a special right triangle with angle measures 30° , 60° , and 90° . Therefore, the length of the hypotenuse, $\overline{DE}$, is 2 times the length of side DF , or $DE = 2(DF)$. Substituting 125 for DF in this equation yields $DE = 2(125)$, or $DE = 250$. Thus, if $\tan(A) = \sqrt{3}$ and $DF = 125$, the length of $\overline{DE}$ is 250.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the length of $\overline{EF}$, not $\overline{DE}$.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$. Circle P has the same center as circle M but has a radius that is twice the radius of circle M . Which equation represents circle P ?

Answer

- A. $(x - 4)^2 + (y - 5)^2 = 18$
- B. $(x - 4)^2 + (y - 5)^2 = 36$
- C. $(x - 8)^2 + (y - 10)^2 = 9$
- D. $(x - 8)^2 + (y - 10)^2 = 18$

Correct Answer: B

Rationale

Choice B is correct. An equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center and r is the radius. It's given that in the xy -plane, circle M is the graph of the equation $(x - 4)^2 + (y - 5)^2 = 9$, which is equivalent to $(x - 4)^2 + (y - 5)^2 = 3^2$. Therefore, the center of circle M is $(4, 5)$ and the radius of circle M is 3 . It's given that circle P has the same center as circle M but has a radius that is twice the radius of circle M . Therefore, the center of circle P is $(4, 5)$ and the radius of circle P is $2(3)$, or 6 . Substituting 4 for h , 5 for k , and 6 for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 4)^2 + (y - 5)^2 = 6^2$, which can be rewritten as $(x - 4)^2 + (y - 5)^2 = 36$.

Choice A is incorrect. The graph of this equation has a radius of $\sqrt{2}(3)$, not $2(3)$.

Choice C is incorrect. The graph of this equation has center $(8, 10)$, not $(4, 5)$, and a radius of 3 , not $2(3)$.

Choice D is incorrect. The graph of this equation has center $(8, 10)$, not $(4, 5)$, and a radius of $\sqrt{2}(3)$, not $2(3)$.

Question ID: 73e1b793

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Right rectangular prism P is similar to right rectangular prism Q. The volume of prism P is **972 cubic centimeters** (**cm³**), and the volume of prism Q is **36 cm³**. One edge of prism P has a length of **12 cm**. What is the length, in **cm**, of the corresponding edge of prism Q?

Correct Answer: 4

Rationale

The correct answer is **4**. It's given that right rectangular prism P is similar to right rectangular prism Q. Therefore, there is a positive constant **k** such that the length of each edge of prism P is **k** times the length of the corresponding edge of prism Q and the volume of prism P is **k³** times the volume of prism Q. It's given that the volume of prism P is **972 cm³** and the volume of prism Q is **36 cm³**. It follows that **972 = 36k³**. Dividing both sides of this equation by **36** yields $\frac{972}{36} = k^3$, or **k³ = 27**. Taking the cube root of both sides of this equation yields **k = 3**. It follows that the length of each edge of prism P is **3** times the length of the corresponding edge of prism Q. It's given that one edge of prism P has a length of **12 cm**. It follows that **12 cm** is **3** times the length of the corresponding edge of prism Q. Thus, the length, in **cm**, of the edge of prism Q corresponding to the edge of prism P with a length of **12 cm** is $\frac{12}{3}$, or **4**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

Correct Answer: 14.66, 14.67, 44/3

Rationale

The correct answer is $\frac{44}{3}$. It's given that in triangle RST , angle T is a right angle. The area of a right triangle can be found using the formula $A = \frac{1}{2}\ell_1\ell_2$, where A represents the area of the right triangle, ℓ_1 represents the length of one leg of the triangle, and ℓ_2 represents the length of the other leg of the triangle. In triangle RST , the two legs are $\overline{RT}$ and $\overline{ST}$. Therefore, if the length of $\overline{RT}$ is 72 and the area of triangle RST is 792, then $792 = \frac{1}{2}(72)(ST)$, or $792 = (36)(ST)$. Dividing both sides of this equation by 36 yields $22 = ST$. Therefore, the length of $\overline{ST}$ is 22. It's also given that point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. It follows that angle LKS is a right angle. Since triangles RST and LSK share angle S and have right angles T and K , respectively, triangles RST and LSK are similar triangles. Therefore, the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is equal to the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$. If the length of $\overline{RT}$ is 72 and the length of $\overline{LK}$ is 24, it follows that the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is $\frac{72}{24}$, or 3, so the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$ is 3. Therefore, $\frac{22}{SK} = 3$. Multiplying both sides of this equation by SK yields $22 = (3)(SK)$. Dividing both sides of this equation by 3 yields $\frac{22}{3} = SK$. Since the length of $\overline{ST}$, 22, is the sum of the length of $\overline{SK}$, $\frac{22}{3}$, and the length of $\overline{KT}$, it follows that the length of $\overline{KT}$ is $22 - \frac{22}{3}$, or $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center O , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 3 inches. What is the circumference, in inches, of the circle?

Answer

- A. 3
- B. 6
- C. 9
- D. 24

Correct Answer: D

Rationale

Choice D is correct. It's given that the measure of arc AB is 45° and the length of arc AB is 3 inches. The arc measure of the full circle is 360° . If x represents the circumference, in inches, of the circle, it follows that $\frac{45^\circ}{360^\circ} = \frac{3 \text{ inches}}{x \text{ inches}}$. This equation is equivalent to $\frac{45}{360} = \frac{3}{x}$, or $\frac{1}{8} = \frac{3}{x}$. Multiplying both sides of this equation by $8x$ yields $1(x) = 3(8)$, or $x = 24$. Therefore, the circumference of the circle is 24 inches.

Choice A is incorrect. This is the length of arc AB .

Choice B is incorrect and may result from multiplying the length of arc AB by 2 .

Choice C is incorrect and may result from squaring the length of arc AB .

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. What is the height, in inches, of this pyramid?

Answer

- A. 56
- B. 90
- C. 106
- D. 180

Correct Answer: A

Rationale

Choice A is correct. The total surface area of a right square pyramid is the sum of the combined surface area of its four lateral faces and the area of its square base. It's given that the right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. It follows that the area of the square base is the difference between the total surface area and the combined surface area of the four lateral faces, or **(70,560 – 38,160)** square inches, which is equal to **32,400** square inches. The area, ***B***, of the base of a square pyramid is given by the equation **$B = s^2$** , where ***s*** is the side length of the base. Substituting **32,400** for ***B*** in this equation yields **$32,400 = s^2$** . Taking the positive square root of both sides of this equation yields **$180 = s$** . The four lateral faces of a square pyramid are congruent triangles. It follows that the area of each lateral face is **$\frac{38,160}{4}$** , or **9,540**, square inches. The area, ***A***, of each lateral face of a square pyramid is given by the equation **Math output error**, where ***s*** is the side length of the pyramid's base and **Math output error** is the pyramid's slant height. Substituting **9,540** for ***A*** and **180** for ***s*** in this equation yields **Math output error**, or **Math output error**. Dividing both sides of this equation by **90** yields **Math output error**. In the right square pyramid, a right triangle can be drawn, where the lengths of the legs correspond to the pyramid's height, ***h***, and half the side length of its base, **$\frac{1}{2}s$** , and the length of the hypotenuse corresponds to the pyramid's slant height, **Math output error**. It follows that the relationship between ***h***, ***s***, and **Math output error** is given by the equation **Math output error**. Substituting **180** for ***s*** and **106** for **Math output error** in this equation yields **$h^2 + ((\frac{1}{2})(180))^2 = (106)^2$** , or **$h^2 + 8,100 = 11,236$** . Subtracting **8,100** from both sides of this equation yields **$h^2 = 3,136$** . Taking the positive square root of both sides of the equation yields ***h* = 56**. Therefore, the height, in inches, of this pyramid is **56**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the pyramid's slant height, not its height, in inches.

Choice D is incorrect. This is the side length of the pyramid's base, not its height, in inches.

Question ID: f6ca90cc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right prism has a square base with side lengths of **10** inches. The height of the prism is **58** inches. What is the volume, in cubic inches, of the prism?

Correct Answer: 5800

Rationale

The correct answer is **5,800**. The volume, V , of a right prism with a square base is given by the formula $V = s^2h$, where s is the side length of the square base and h is the height of the prism. It's given that a right prism has a square base with side lengths of **10** inches. It's also given that the height of the prism is **58** inches. Substituting **10** for s and **58** for h in the formula $V = s^2h$ yields $V = (10)^2(58)$, or $V = 5,800$. Therefore, the volume, in cubic inches, of the prism is **5,800**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$(x - 6)^2 + (y + 5)^2 = 16$$

In the xy-plane, the graph of the equation above is a circle. Point P is on the circle and has coordinates $(10, -5)$. If $\overline{PQ}$ is a diameter of the circle, what are the coordinates of point Q ?

- Answer**
- A. $(2, -5)$
- B. $(6, -1)$
- C. $(6, -5)$
- D. $(6, -9)$

Correct Answer: A

Rationale

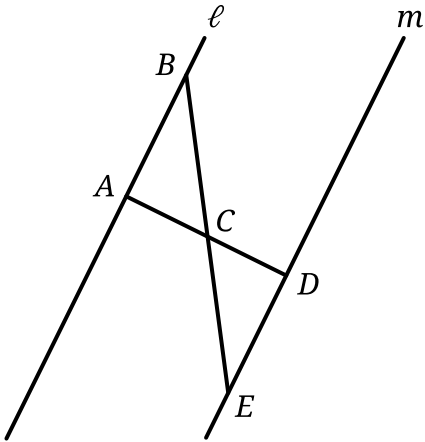
Choice A is correct. The standard form for the equation of a circle is $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) are the coordinates of the center and r is the length of the radius. According to the given equation, the center of the circle is $(6, -5)$. Let (x_1, y_1) represent the coordinates of point Q. Since point P $(10, -5)$ and point Q (x_1, y_1) are the endpoints of a diameter of the circle, the center $(6, -5)$ lies on the diameter, halfway between P and Q. Therefore, the following relationships hold: $\frac{x_1 + 10}{2} = 6$ and $\frac{y_1 + (-5)}{2} = -5$. Solving the equations for x_1 and y_1 , respectively, yields $x_1 = 2$ and $y_1 = -5$. Therefore, the coordinates of point Q are $(2, -5)$.

Alternate approach: Since point P $(10, -5)$ on the circle and the center of the circle $(6, -5)$ have the same y-coordinate, it follows that the radius of the circle is $10 - 6 = 4$. In addition, the opposite end of the diameter $\overline{PQ}$ must have the same y-coordinate as P and be 4 units away from the center. Hence, the coordinates of point Q must be $(2, -5)$.

Choices B and D are incorrect because the points given in these choices lie on a diameter that is perpendicular to the diameter $\overline{PQ}$. If either of these points were point Q, then $\overline{PQ}$ would not be the diameter of the circle. Choice C is incorrect because $(6, -5)$ is the center of the circle and does not lie on the circle.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, segment AD and segment BE intersect at point C . Segment AD is perpendicular to line ℓ at point A . Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC ?

- I. Segment AD is perpendicular to line m at point D .
- II. $AC = 13$ and $DC = 13$

Answer

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

Correct Answer: D

Rationale

Choice D is correct. In the figure, it's given that segment AD is perpendicular to line ℓ at point A . Statement I is that segment AD is perpendicular to line m at point D . It follows that angles BAC and EDC are right angles. Therefore, triangles ABC and DEC are right triangles. At least one pair of congruent corresponding sides is needed to determine triangle congruence by the side-angle-side (SAS) condition or the angle-side-angle (ASA) condition, so statement I is not sufficient to determine that triangle ABC is congruent to triangle DEC . Statement II is that $AC = 13$ and $DC = 13$. It follows that triangles ABC and DEC share one pair of congruent corresponding sides. Two additional pairs of congruent corresponding parts are needed to determine triangle congruence by SAS condition, ASA condition, or side-side-side (SSS) condition, so statement II is not sufficient to determine that triangle ABC is congruent to triangle DEC . Therefore, neither I nor II is sufficient by itself to determine that triangle ABC is congruent to triangle DEC .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 89661424

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has its center at $(-5, 2)$ and has a radius of 9 . An equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. What is the value of c ?

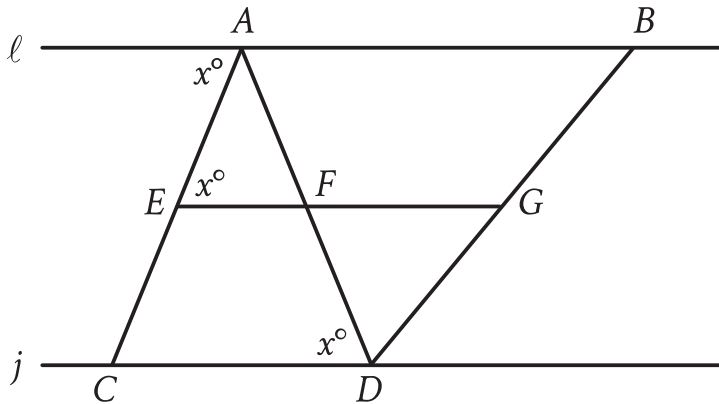
Correct Answer: -52

Rationale

The correct answer is -52 . The equation of a circle in the xy -plane with its center at (h, k) and a radius of r can be written in the form $(x - h)^2 + (y - k)^2 = r^2$. It's given that a circle in the xy -plane has its center at $(-5, 2)$ and has a radius of 9 . Substituting -5 for h , 2 for k , and 9 for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - (-5))^2 + (y - 2)^2 = 9^2$, or $(x + 5)^2 + (y - 2)^2 = 81$. It's also given that an equation of this circle is $x^2 + y^2 + ax + by + c = 0$, where a , b , and c are constants. Therefore, $(x + 5)^2 + (y - 2)^2 = 81$ can be rewritten in the form $x^2 + y^2 + ax + by + c = 0$. The equation $(x + 5)^2 + (y - 2)^2 = 81$, or $(x + 5)(x + 5) + (y - 2)(y - 2) = 81$, can be rewritten as $x^2 + 5x + 5x + 25 + y^2 - 2y - 2y + 4 = 81$. Combining like terms on the left-hand side of this equation yields $x^2 + y^2 + 10x - 4y + 29 = 81$. Subtracting 81 from both sides of this equation yields $x^2 + y^2 + 10x - 4y - 52 = 0$, which is equivalent to $x^2 + y^2 + 10x + (-4)y + (-52) = 0$. This equation is in the form $x^2 + y^2 + ax + by + c = 0$. Therefore, the value of c is -52 .

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F , and line **Math output error** is parallel to line j . If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

- I. $AB = 2.25EF$
- II. $FG = 0.5AB$
- III. $\frac{EF}{FG} = \frac{AF}{FD}$

Answer

- A. I only
- B. II only
- C. I and II
- D. II and III

Correct Answer: B

Rationale

Choice B is correct. In the figure, $\overline{AE}$ intersects line **Math output error** and $\overline{EG}$, and alternate interior angles are both marked x° . It follows that $\overline{EG}$ is parallel to line **Math output error**. It's given that line **Math output error** is parallel to line j , so $\overline{EG}$ is also parallel to line j . Thus, the following angles also have measure x° : $\angle EFA$, $\angle ACD$, $\angle BAD$, and $\angle GFD$. Triangles AEF and ACD each have two angles with measure x° , so triangles AEF and ACD are similar. It's also given that $EF = 0.5CD$, so $AF = 0.5AD$. It follows that $AF = FD$. Triangles DFG and DAB each have one angle with measure x° and share $\angle FDG$, so triangles DFG and DAB are similar. Since $AF = FD$, it follows that $AD = 2FD$, so $AB = 2FG$. Substituting $2FG$ for AB in statement I yields $2FG = 2.25EF$. It's given that $FG = 2.25CD$. Substituting $2.25CD$ for FG and $0.5CD$ for EF in the equation $2FG = 2.25EF$ yields $2(2.25CD) = 2.25(0.5CD)$, or $4.5CD = 1.125CD$. Thus, statement I is not true. Since $AB = 2FG$, it follows that $0.5AB = FG$, or $FG = 0.5AB$. Therefore, statement II must be true. Substituting $0.5CD$ for EF , $2.25CD$ for FG , and FD for AF in statement III yields $\frac{0.5CD}{2.25CD} = \frac{FD}{FD}$, or $\frac{2}{9} = 1$. Thus, statement III is not true. Therefore, statement II only must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 7c25b0dc

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

The length of a rectangle’s diagonal is $3\sqrt{17}$, and the length of the rectangle’s shorter side is 3 . What is the length of the rectangle’s longer side?

Correct Answer: 12

Rationale

The correct answer is 12 . The diagonal of a rectangle forms a right triangle, where the shorter side and the longer side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. It's given that the length of the rectangle's diagonal is $3\sqrt{17}$ and the length of the rectangle's shorter side is 3 . Thus, the length of the hypotenuse of the right triangle formed by the diagonal is $3\sqrt{17}$ and the length of one of the legs is 3 . By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting $3\sqrt{17}$ for c and 3 for b in this equation yields $a^2 + (3)^2 = (3\sqrt{17})^2$, or $a^2 + 9 = 153$. Subtracting 9 from both sides of this equation yields $a^2 = 144$. Taking the square root of both sides of this equation yields $a = \pm\sqrt{144}$, or $a = \pm 12$. Since a represents a length, which must be positive, the value of a is 12 . Thus, the length of the rectangle's longer side is 12 .

Question ID: 4fb972f2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS ?

Correct Answer: 36

Rationale

The correct answer is **36**. It's given that side SQ is the hypotenuse of right triangle QRS . It's also given that the length of side SQ is $4\sqrt{82}$ and the length of side QR is 4. It follows that sides QR and RS are the legs of triangle QRS . The Pythagorean theorem states that for a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Let b represent the length of side RS . By the Pythagorean theorem, $4^2 + b^2 = (4\sqrt{82})^2$, or $16 + b^2 = 1,312$. Subtracting 16 from both sides of this equation yields $b^2 = 1,296$. Taking the positive square root of both sides of this equation yields $b = 36$. Therefore, the length of side RS is 36.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

Answer

- A. 11
- B. 22
- C. 44
- D. 66

Correct Answer: C

Rationale

Choice C is correct. The volume of right circular cylinder A is given by the expression $\pi r^2 h$, where r is the radius of its circular base and h is its height. The volume of a cylinder with twice the radius and half the height of cylinder A is given by $\pi (2r)^2 \left(\frac{1}{2}h\right)$, which is equivalent to $4\pi r^2 \left(\frac{1}{2}h\right) = 2\pi r^2 h$. Therefore, the volume is twice the volume of cylinder A, or $2 \times 22 = 44$.

Choice A is incorrect and likely results from not multiplying the radius of cylinder A by 2. Choice B is incorrect and likely results from not squaring the 2 in 2r when applying the volume formula. Choice D is incorrect and likely results from a conceptual error.

Question ID: 149021da

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center P , and points A and B lie on the circle. The measure of arc AB is 45° and the length of arc AB is 4π units. What is the length, in units, of the radius of the circle?

Correct Answer: 16

Rationale

The correct answer is **16**. It's given that the measure of arc AB is 45° and the length of arc AB is 4π units. In a circle, the circumference corresponds to an arc measure of 360° . Therefore, the ratio of the measure of arc AB to 360° is equal to the ratio of the length of arc AB to the circumference. Since the circumference of a circle is $2\pi r$, it follows that $\frac{45}{360} = \frac{4\pi}{2\pi r}$. This equation can be rewritten as $\frac{1}{8} = \frac{2}{r}$. Multiplying both sides of this equation by $8r$ yields $1(r) = 2(8)$, or $r = 16$. Therefore, the length, in units, of the radius of the circle is **16**.

Question ID: fb58c0db

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

Points A and B lie on a circle with radius 1, and arc $\overline{AB}$ has length $\frac{\pi}{3}$. What fraction of the circumference of the circle is the length of arc $\overline{AB}$?

Rationale

The correct answer is $\frac{1}{6}$. The circumference, C, of a circle is $C = 2\pi r$, where r is the length of the radius of the circle. For the given circle with a radius of 1, the circumference is $C = 2(\pi)(1)$, or $C = 2\pi$. To find what fraction of the circumference the length of arc $\overline{AB}$ is, divide the length of the arc by the circumference, which gives $\frac{\pi}{3} \div 2\pi$. This division can be represented by $\frac{\pi}{3} \cdot \frac{1}{2\pi} = \frac{1}{6}$. Note that 1/6, .1666, .1667, 0.166, and 0.167 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. The length of $\overline{AB}$ is 2.9 times the length of $\overline{DE}$. If $\tan A = \frac{21}{20}$, what is the value of $\sin D$?

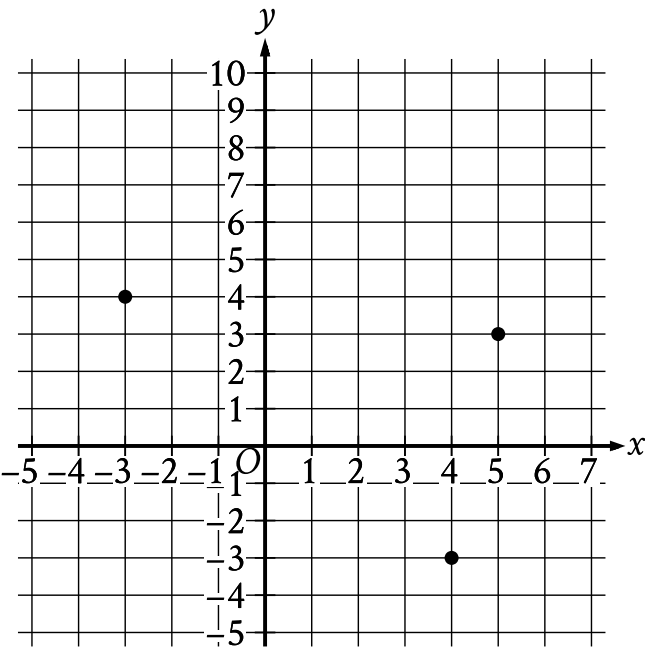
Correct Answer: .7241, 21/29

Rationale

The correct answer is $\frac{21}{29}$. It's given that triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angles C and F are right angles. In similar triangles, the tangents of corresponding angles are equal. Therefore, if $\tan A = \frac{21}{20}$, then $\tan D = \frac{21}{20}$. In a right triangle, the tangent of an acute angle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Therefore, in triangle DEF , if $\tan D = \frac{21}{20}$, the ratio of the length of $\overline{EF}$ to the length of $\overline{DF}$ is $\frac{21}{20}$. If the lengths of $\overline{EF}$ and $\overline{DF}$ are 21 and 20, respectively, then the ratio of the length of $\overline{EF}$ to the length of $\overline{DF}$ is $\frac{21}{20}$. In a right triangle, the sine of an acute angle is the ratio of the length of the leg opposite the angle to the length of the hypotenuse. Therefore, the value of $\sin D$ is the ratio of the length of $\overline{EF}$ to the length of $\overline{DE}$. The length of $\overline{DE}$ can be calculated using the Pythagorean theorem, which states that if the lengths of the legs of a right triangle are a and b and the length of the hypotenuse is c , then $a^2 + b^2 = c^2$. Therefore, if the lengths of $\overline{EF}$ and $\overline{DF}$ are 21 and 20, respectively, then $(21)^2 + (20)^2 = (DE)^2$, or $841 = (DE)^2$. Taking the positive square root of both sides of this equation yields $29 = DE$. Therefore, if the lengths of $\overline{EF}$ and $\overline{DF}$ are 21 and 20, respectively, then the length of $\overline{DE}$ is 29 and the ratio of the length of $\overline{EF}$ to the length of $\overline{DE}$ is $\frac{21}{29}$. Thus, if $\tan A = \frac{21}{20}$, the value of $\sin D$ is $\frac{21}{29}$. Note that 21/29, .7241, and 0.724 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question



What is the area, in square units, of the triangle formed by connecting the three points shown?

Correct Answer: 24.5, 49/2

Rationale

The correct answer is **24.5**. It's given that a triangle is formed by connecting the three points shown, which are $(-3, 4)$, $(5, 3)$, and $(4, -3)$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $(-3, 4)$, $(5, 4)$, $(5, -3)$, and $(-3, -3)$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(5, -3)$. This distance is $4 - (-3)$, or 7. The length, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or 8. It follows that the area, in square units, of the rectangle is $(7)(8)$, or 56. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $(-3, 4)$, $(5, 4)$, and $(5, 3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(5, 3)$. This distance is $4 - 3$, or 1. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or 8. It follows that the area, in square units, of this triangle is $\frac{1}{2}(8)(1)$, or 4. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $(4, -3)$, $(5, 3)$, and $(5, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 3)$ and $(5, -3)$. This distance is $3 - (-3)$, or 6. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, -3)$ and $(4, -3)$. This distance is $5 - 4$, or 1. It follows that the area, in square units, of this triangle is $\frac{1}{2}(1)(6)$, or 3. The third triangle that lies inside the rectangle but outside triangle A is formed by the points $(-3, 4)$, $(-3, -3)$, and $(4, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(4, -3)$ and $(-3, -3)$. This distance is $4 - (-3)$, or 7. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(-3, 4)$ and $(-3, -3)$. This distance is $4 - (-3)$, or 7. It follows that the area, in square units, of this triangle is $\frac{1}{2}(7)(7)$, or 24.5. Thus, the area, in square

units, of the triangle formed by connecting the three points shown is **56 — 4 — 3 — 24.5**, or **24.5**. Note that 24.5 and 49/2 are examples of ways to enter a correct answer.

Question ID: 4757123b

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Correct Answer: 1681

Rationale

The correct answer is **1,681**. It's given that trapezoid A and trapezoid B are similar. When two trapezoids are similar, the scale factor between their corresponding side lengths is ***k***, and the scale factor between their areas is ***k*²**. It's given that the length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B, so the scale factor between their corresponding side lengths is **41**. It follows that the scale factor between the areas is **41²**, or **1,681**. Therefore, the area of trapezoid A is **1,681** times as large as the area of trapezoid B.

Question ID: 92eb236a

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In a right triangle, the tangent of one of the two acute angles is $\frac{\sqrt{3}}{3}$. What is the tangent of the other acute angle?

Answer

- A. $-\frac{\sqrt{3}}{3}$
- B. $-\frac{3}{\sqrt{3}}$
- C. $\frac{\sqrt{3}}{3}$
- D. $\frac{3}{\sqrt{3}}$

Correct Answer: D

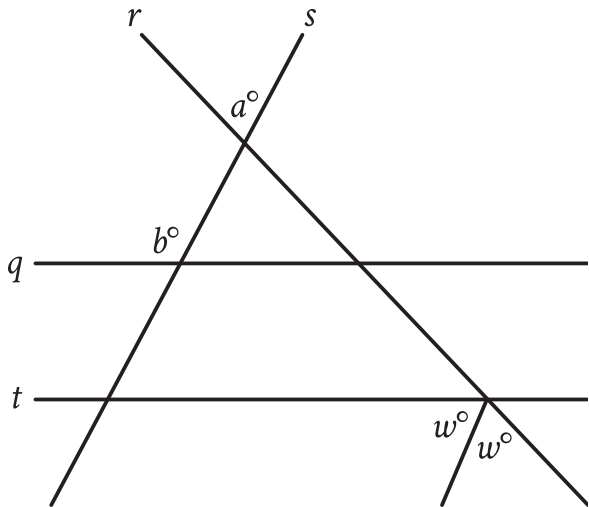
Rationale

Choice D is correct. The tangent of a nonright angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. Using that definition for tangent, in a right triangle with legs that have lengths a and b, the tangent of one acute angle is $\frac{a}{b}$ and the tangent for the other acute angle is $\frac{b}{a}$. It follows that the tangents of the acute angles in a right triangle are reciprocals of each other. Therefore, the tangent of the other acute angle in the given triangle is the reciprocal of $\frac{\sqrt{3}}{3}$ or $\frac{3}{\sqrt{3}}$.

Choice A is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the tangent of the angle described. Choice B is incorrect and may result from assuming that the tangent of the other acute angle is the negative of the reciprocal of the tangent of the angle described. Choice C is incorrect and may result from interpreting the tangent of the other acute angle as equal to the tangent of the angle described.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

Correct Answer: 101/2, 50.5

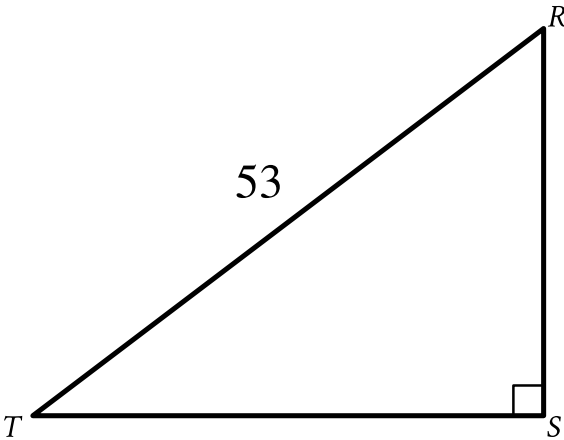
Rationale

The correct answer is $\frac{101}{2}$. In the figure, lines q , r , and s form a triangle. One interior angle of this triangle is vertical to the angle marked a° ; therefore, the interior angle also has measure a° . It's given that $a = 43$. Therefore, the interior angle of the triangle has measure 43° . A second interior angle of the triangle forms a straight line, q , with the angle marked b° . Therefore, the sum of the measures of these two angles is 180° . It's given that $b = 122$. Therefore, the angle marked b° has measure 122° and the second interior angle of the triangle has measure $(180 - 122)^\circ$, or 58° . The sum of the interior angles of a triangle is 180° . Therefore, the measure of the third interior angle of the triangle is $(180 - 43 - 58)^\circ$, or 79° . It's given that parallel lines q and t are intersected by line r . It follows that the triangle's interior angle with measure 79° is congruent to the same side interior angle between lines q and t formed by lines t and r . Since this angle is supplementary to the two angles marked w° , the sum of 79° , w° , and w° is 180° . It follows that $79 + w + w = 180$, or $79 + 2w = 180$. Subtracting 79 from both sides of this equation yields $2w = 101$. Dividing both sides of this equation by 2 yields $w = \frac{101}{2}$. Note that 101/2 and 50.5 are examples of ways to enter a correct answer.

Question ID: a67b9f88

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the triangle shown, $RS = \sqrt{105}$. What is the value of $\sin R$?

Correct Answer: .9811, 52/53

Rationale

The correct answer is $\frac{52}{53}$. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. The length of the hypotenuse of the right triangle shown is 53. It's given that $RS = \sqrt{105}$. Therefore, the length of one of the legs of the triangle shown is $\sqrt{105}$. Let x represent TS , the length of the other leg of the triangle shown. Therefore, $53^2 = (\sqrt{105})^2 + x^2$, or $2,809 = 105 + x^2$. Subtracting 105 from both sides of this equation yields $2,704 = x^2$. Taking the positive square root of both sides of this equation yields $52 = x$. Therefore, TS , the length of the other leg of the triangle shown, is 52. The sine of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the hypotenuse. The length of the leg opposite angle R is 52, and the length of the hypotenuse is 53. Therefore, the value of $\sin R$ is $\frac{52}{53}$. Note that 52/53 or .9811 are examples of ways to enter a correct answer.

Question ID: f7e626b2

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

Answer

- A. 30
- B. 74
- C. 120
- D. 148

Rationale

Choice D is correct. The surface area is found by summing the area of each face. A right rectangular prism consists of three pairs of congruent rectangles, so the surface area is found by multiplying the areas of three adjacent rectangles by 2 and adding these products. For this prism, the surface area is equal to $2(4 \cdot 5) + 2(5 \cdot 6) + 2(4 \cdot 6)$, or $2(20) + 2(30) + 2(24)$, which is equal to 148.

Choice A is incorrect. This is the area of one of the faces of the prism. Choice B is incorrect and may result from adding the areas of three adjacent rectangles without multiplying by 2. Choice C is incorrect. This is the volume, in cubic inches, of the prism.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angle C corresponds to angle F . Angles C and F are right angles. If $\tan(A) = \frac{50}{7}$, what is the value of $\tan(E)$?

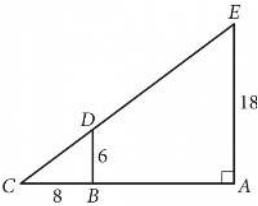
Correct Answer: .14, 7/50

Rationale

The correct answer is $\frac{7}{50}$. It's given that triangle ABC is similar to triangle DEF , where angle A corresponds to angle D and angle C corresponds to angle F . In similar triangles, the tangents of corresponding angles are equal. Since angle A and angle D are corresponding angles, if $\tan(A) = \frac{50}{7}$, then $\tan(D) = \frac{50}{7}$. It's also given that angles C and F are right angles. It follows that triangle DEF is a right triangle with acute angles D and E . The tangent of one acute angle in a right triangle is the inverse of the tangent of the other acute angle in the triangle. Therefore, $\tan(E) = \frac{1}{\tan(D)}$. Substituting $\frac{50}{7}$ for $\tan(D)$ in this equation yields $\tan(E) = \frac{1}{\frac{50}{7}}$, or $\tan(E) = \frac{7}{50}$. Thus, if $\tan(A) = \frac{50}{7}$, the value of $\tan(E)$ is $\frac{7}{50}$. Note that 7/50 and .14 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the figure above, $\overline{BD}$ is parallel to $\overline{AE}$. What is the length of $\overline{CE}$?

Rationale

The correct answer is 30. In the figure given, since $\overline{BD}$ is parallel to $\overline{AE}$ and both segments are intersected by $\overline{CE}$, then angle BDC and angle AEC are corresponding angles and therefore congruent. Angle BCD and angle ACE are also congruent because they are the same angle. Triangle BCD and triangle ACE are similar because if two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. Since triangle BCD and triangle ACE are similar, their corresponding sides are proportional. So in triangle BCD and triangle ACE, $\overline{BD}$ corresponds to $\overline{AE}$ and $\overline{CD}$ corresponds to $\overline{CE}$. Therefore, $\frac{BD}{CD} = \frac{AE}{CE}$. Since triangle BCD is a right triangle, the Pythagorean theorem can be used to give the value of CD: $6^2 + 8^2 = CD^2$. Taking the square root of each side gives $CD = 10$. Substituting the values in the proportion $\frac{BD}{CD} = \frac{AE}{CE}$ yields $\frac{6}{10} = \frac{18}{CE}$. Multiplying each side by CE, and then multiplying by $\frac{10}{6}$ yields $CE = 30$. Therefore, the length of $\overline{CE}$ is 30.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle in the xy -plane has equation $(x + 3)^2 + (y - 1)^2 = 25$. Which of the following points does NOT lie in the interior of the circle?

Answer

- A. $(-7, 3)$
- B. $(-3, 1)$
- C. $(0, 0)$
- D. $(3, 2)$

Correct Answer: D

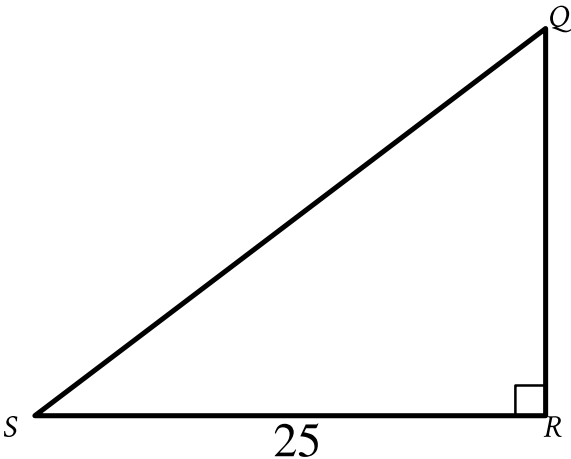
Rationale

Choice D is correct. The circle with equation $(x + 3)^2 + (y - 1)^2 = 25$ has center $(-3, 1)$ and radius 5. For a point to be inside of the circle, the distance from that point to the center must be less than the radius, 5. The distance between $(3, 2)$ and $(-3, 1)$ is $\sqrt{(-3 - 3)^2 + (1 - 2)^2} = \sqrt{(-6)^2 + (-1)^2} = \sqrt{37}$, which is greater than 5. Therefore, $(3, 2)$ does NOT lie in the interior of the circle.

Choice A is incorrect. The distance between $(-7, 3)$ and $(-3, 1)$ is $\sqrt{(-7 + 3)^2 + (3 - 1)^2} = \sqrt{(-4)^2 + (2)^2} = \sqrt{20}$, which is less than 5, and therefore $(-7, 3)$ lies in the interior of the circle. Choice B is incorrect because it is the center of the circle. Choice C is incorrect because the distance between $(0, 0)$ and $(-3, 1)$ is $\sqrt{(0 + 3)^2 + (0 - 1)^2} = \sqrt{(3)^2 + (1)^2} = \sqrt{8}$, which is less than 5, and therefore $(0, 0)$ in the interior of the circle.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- Answer
- A. $25 \cos Q$
 - B. $25 \sin Q$
 - C. $\frac{25}{\cos Q}$
 - D. $\frac{25}{\sin Q}$

Correct Answer: D

Rationale

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 25. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 25 for RS in this equation, which yields $\sin Q = \frac{25}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 25$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{25}{\sin Q}$. Therefore, the expression $\frac{25}{\sin Q}$ represents the length of $\overline{QS}$.

Choice A is incorrect and may result from conceptual or calculation errors.

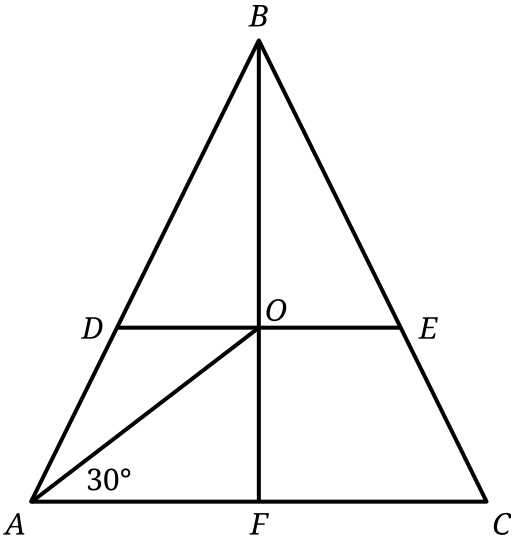
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 1dacfb94

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In triangle ABC , $AB = BC$. Points D and E lie on line segments AB and BC , respectively, such that line segments DE and AC are parallel. Point O is the midpoint of line segment DE , and the measure of $\angle OAC$ is 30° . Point F lies on line segment AC , and line segment BF passes through point O . If $AB = 81$ and $AO = BO$, what is the length of line segment BE ?

Correct Answer: 54

Rationale

The correct answer is 54. It's given that $AB = BC$, so triangle ABC is isosceles. It's also given that line segments DE and AC are parallel. Since line segments DE and AC are parallel, triangle BDE is similar to triangle BAC , which means triangle BDE is also isosceles. It's given that point O is the midpoint of line segment DE . Since triangle BDE is isosceles, line segment BO splits triangle BDE into two congruent triangles, where $\angle BOD$ is a right angle. Therefore, $\angle BFA$ is also a right angle. The figure shows that the measure of $\angle OAF$ is 30° . It follows that triangle AOF is a 30-60-90 triangle, where the measure of $\angle AOF$ is 60° . In a 30-60-90 triangle, the side opposite the 30° angle is half the length of the side opposite the 90° angle. Let x represent the length of line segment OF and let $2x$ represent the length of line segment AO . It's given that $AO = BO$. Since $AO = BO$, the length of line segment BO is also $2x$. It follows that the length of line segment BF is $BO + OF$, which is $2x + x$, or $3x$. Since triangle BDE is similar to triangle BAC , the ratio of corresponding side lengths is equal to the ratio of the heights of the triangles. This means that $\frac{BE}{BC} = \frac{BO}{BF}$. Substituting $2x$ for BO and $3x$ for BF in this equation yields $\frac{BE}{BC} = \frac{2x}{3x}$, or $\frac{BE}{BC} = \frac{2}{3}$. It's given that $AB = 81$ and $AB = BC$, so $BC = 81$. Substituting 81 for BC in the equation $\frac{BE}{BC} = \frac{2}{3}$ yields $\frac{BE}{81} = \frac{2}{3}$. Multiplying both sides of this equation by 81 yields $BE = \frac{2}{3}(81)$, or $BE = 54$.

Question ID: 10e059a6

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle A is $\frac{\pi}{216}$ radians. The measure of angle B is 36 times the measure of angle A . What is the value of $\cos B$?

Answer

- A. 0
- B. $\frac{1}{2}$
- C. $\frac{\sqrt{2}}{2}$
- D. $\frac{\sqrt{3}}{2}$

Correct Answer: D

Rationale

Choice D is correct. It’s given that the measure of angle A is $\frac{\pi}{216}$ radians and the measure of angle B is 36 times the measure of angle A . Therefore, the measure of angle B is $36\left(\frac{\pi}{216}\right)$ radians, or $\frac{\pi}{6}$ radians. A trigonometric ratio can be found using the unit circle, or a circle with radius 1 unit. For a unit circle centered at the origin in the xy -plane, the cosine of $\frac{\pi}{6}$ radians is the x -coordinate of the point resulting from traveling a distance of $\frac{\pi}{6}$ counterclockwise from the point $(1, 0)$. This results in the point $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$. Therefore, the value of $\cos B$ is $\frac{\sqrt{3}}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\sin B$, not $\cos B$.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 9f9112ab

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

The measure of angle L is **60** degrees. If the measure of angle L is $\frac{\pi}{n}$ radians, where n is a constant, what is the value of n ?

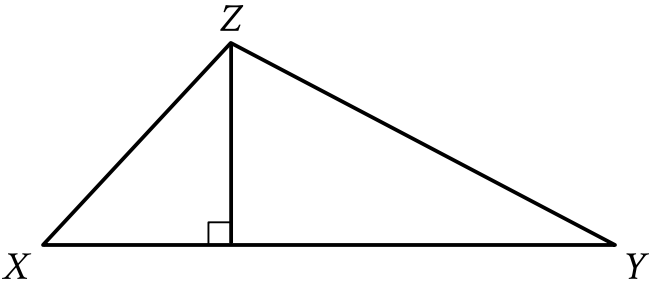
Correct Answer: 3

Rationale

The correct answer is **3**. The measure of an angle in radians is equal to the measure of the angle in degrees multiplied by $\frac{\pi}{180}$. It's given that the measure of angle L is **60** degrees. Multiplying **60** by $\frac{\pi}{180}$ yields $\frac{60\pi}{180}$, or $\frac{\pi}{3}$. It's also given that the measure of angle L is $\frac{\pi}{n}$ radians, so $\frac{\pi}{n} = \frac{\pi}{3}$. Dividing both sides of this equation by π yields $\frac{1}{n} = \frac{1}{3}$. Taking the reciprocal of both sides of this equation yields $\frac{n}{1} = \frac{3}{1}$, or $n = 3$. Therefore, the value of n is **3**.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

- Answer
- A. 247
 - B. 494
 - C. $247 \sin 54^\circ$
 - D. $494 \sin 54^\circ$

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of a base of the triangle and h is its corresponding height. It's given that the length of $\overline{XY}$ is 26 units. This length represents a base of the triangle, so $b = 26$. Let h represent its corresponding height. It's given that the measure of angle X is 54° and the length of $\overline{XZ}$ is 19 units. It follows that $\sin 54^\circ = \frac{h}{19}$. Multiplying both sides of this equation by 19 yields $19 \sin 54^\circ = h$. Substituting 26 for b and $19 \sin 54^\circ$ for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(26)(19 \sin 54^\circ)$, or $A = 247 \sin 54^\circ$. Therefore, the area, in square units, of triangle XYZ is $247 \sin 54^\circ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

In the xy -plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$. Circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = p$, where p is a constant. What is the value of p ?

Correct Answer: 49

Rationale

The correct answer is 49. An equation of a circle in the xy -plane can be written as $(x - h)^2 + (y - k)^2 = r^2$, where the center of the circle is (h, k) and the radius of the circle is r . It's given that in the xy -plane, circle F is defined by the equation $(x + 8)^2 + (y + 8)^2 = 25$, which can be rewritten as $(x - (-8))^2 + (y - (-8))^2 = 5^2$. It follows that $h = -8$, $k = -8$, and $r = 5$. Therefore, the center of circle F is $(-8, -8)$ and the radius of circle F is 5. It's given that circle G has the same center as circle F , and the radius of circle G is 2 units greater than the radius of circle F . Therefore, the center of circle G is $(-8, -8)$ and the radius of circle G is $5 + 2$, or 7. Substituting -8 for h , -8 for k , and 7 for r into the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x + 8)^2 + (y + 8)^2 = 7^2$, which is equivalent to $(x + 8)^2 + (y + 8)^2 = 49$. It follows that circle G is defined by the equation $(x + 8)^2 + (y + 8)^2 = 49$; therefore, the value of p is 49.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x ?

Answer

- A. 5
- B. 19
- C. 23
- D. 29

Correct Answer: A

Rationale

Choice A is correct. It's given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other angle, the angles are complementary. It follows that $\angle Q$ and $\angle R$ are complementary. That is, the sum of the measures of the angles is 90 degrees. It's given that the measure of $\angle Q$ is $x + 61$ degrees and the measure of $\angle R$ is $4x + 4$ degrees. It follows that $(x + 61) + (4x + 4) = 90$. By combining like terms, this equation can be rewritten as $5x + 65 = 90$. Subtracting 65 from each side of this equation yields $5x = 25$. Dividing each side of this equation by 5 yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\cos(Q) = \cos(R)$ rather than $\cos(Q) = \sin(R)$.

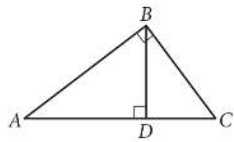
Choice C is incorrect. This would be the value of x if $\cos(Q) = -\cos(R)$ rather than $\cos(Q) = \sin(R)$ and if $\angle R$ were obtuse rather than acute.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 6a3fbec3

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$. What is the length of $\overline{DC}$?

Rationale

The correct answer is 4.5. According to the properties of right triangles, BD divides triangle ABC into two similar triangles, ABD and BCD. The corresponding sides of ABD and BCD are proportional, so the ratio of BD to AD is the same as the ratio of DC to BD. Expressing this information as a proportion gives $\frac{6}{8} = \frac{DC}{6}$. Solving the proportion for DC results in $DC = 4.5$. Note that 4.5 and 9/2 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

A circle has center O , and points A and U lie on the circle. Line segments AE and UE are tangent to the circle at points A and U , respectively. The distance between point O and point E is 510 centimeters and the distance between point O and point U is 78 centimeters. What is the area, in square centimeters, of quadrilateral $OAEU$?

Correct Answer: 39312

Rationale

The correct answer is **39,312**. It's given that the circle has center O and points A and U lie on the circle. It follows that segments OA and OU are radii of the circle and therefore congruent. It's also given that line segments AE and UE are tangent to the circle at points A and U , respectively. Since line segments AE and UE are tangent to the circle from a common external point, point E , line segments AE and UE are congruent and form right angles with radii OA and OU , respectively. Therefore, quadrilateral $OAEU$ consists of two congruent right triangles: triangle OAE and triangle OUE . It's given that the distance between point O and point E is 510 centimeters and the distance between point O and point U is 78 centimeters. It follows that the area of triangle OUE can be found by first using the Pythagorean theorem to calculate the distance between point U and point E . For right triangle OUE , where angle U is a right angle, the Pythagorean theorem states that $(OU)^2 + (UE)^2 = (OE)^2$. Substituting 78 for OU and 510 for OE in this equation yields $78^2 + (UE)^2 = 510^2$, or $6,084 + (UE)^2 = 260,100$. Subtracting 6,084 from both sides of this equation yields $(UE)^2 = 254,016$. Taking the positive square root of both sides of this equation yields $UE = 504$. Thus, triangle OUE is a right triangle with legs of length 78 centimeters and 504 centimeters. It follows that the area of triangle OUE is $(\frac{1}{2})(78)(504)$ square centimeters, or 19,656 square centimeters. Since triangle OAE is congruent to triangle OUE , the area of triangle OAE is also 19,656 square centimeters. Therefore, the area of quadrilateral $OAEU$ is $19,656 + 19,656$ square centimeters, or **39,312** square centimeters.

Question ID: 459dd6c5

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF ?

Correct Answer: 135/8, 16.87, 16.88

Rationale

The correct answer is $\frac{135}{8}$. It's given that triangles ABC and DEF are similar and each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . For two similar triangles, if each side length of the first triangle is k times the corresponding side length of the second triangle, then the area of the first triangle is k^2 times the area of the second triangle. Therefore, the area of triangle ABC is 4^2 , or 16 , times the area of triangle DEF . It's given that the area of triangle ABC is 270 square inches. Let a represent the area, in square inches, of triangle DEF . It follows that 270 is 16 times a , or $270 = 16a$. Dividing both sides of this equation by 16 yields $\frac{270}{16} = a$, which is equivalent to $\frac{135}{8} = a$. Thus, the area, in square inches, of triangle DEF is $\frac{135}{8}$. Note that 135/8, 16.87, and 16.88 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Circles	Hard

Question

$$x^2 + y^2 - 10x - 8y - 14n = 0$$

The given equation represents circle A in the xy -plane, where n is a constant. Point $(11, 8)$ lies on circle B, which has the same center but twice the diameter as circle A. What is the value of n ?

Correct Answer: -2

Rationale

The correct answer is -2 . It's given that $x^2 + y^2 - 10x - 8y - 14n = 0$ represents circle A in the xy -plane, where n is a constant. The equation of a circle in the xy -plane can be written in the form $(x - h)^2 + (y - k)^2 = r^2$, where (h, k) is the center of the circle and r is its radius. By completing the square, the equation $x^2 + y^2 - 10x - 8y - 14n = 0$ can be rewritten as $\left(x^2 - 10x + \left(\frac{-10}{2}\right)^2\right) + \left(y^2 - 8y + \left(\frac{-8}{2}\right)^2\right) = 14n + \left(\frac{-10}{2}\right)^2 + \left(\frac{-8}{2}\right)^2$. This is equivalent to $(x^2 - 10x + 25) + (y^2 - 8y + 16) = 14n + 25 + 16$, which can be written as $(x - 5)^2 + (y - 4)^2 = 14n + 41$. It follows that $(5, 4)$ is the center of circle A and $\sqrt{14n + 41}$ is its radius. It's given that circle B has the same center as circle A. Therefore, the center of circle B is $(5, 4)$. It's also given that circle B has twice the diameter as circle A. Therefore, circle B has twice the radius as circle A. Since the radius of circle A is $\sqrt{14n + 41}$, it follows that the radius of circle B is $2\sqrt{14n + 41}$. Substituting 5 for h , 4 for k , and $2\sqrt{14n + 41}$ for r in the equation $(x - h)^2 + (y - k)^2 = r^2$ yields $(x - 5)^2 + (y - 4)^2 = \left(2\sqrt{14n + 41}\right)^2$, or $(x - 5)^2 + (y - 4)^2 = 4(14n + 41)$, which represents an equation for circle B. It's given that the point $(11, 8)$ lies on circle B. Substituting 11 for x and 8 for y in the equation $(x - 5)^2 + (y - 4)^2 = 4(14n + 41)$ yields $(11 - 5)^2 + (8 - 4)^2 = 4(14n + 41)$, which gives $6^2 + 4^2 = 4(14n + 41)$, or $52 = 4(14n + 41)$. Dividing both sides of this equation by 4 yields $13 = 14n + 41$. Subtracting 41 from both sides of this equation yields $-28 = 14n$. Dividing both sides of this equation by 14 yields $-2 = n$. Therefore, the value of n is -2 .

Question ID: 25da87f8

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18 + 6\sqrt{3}$. What is the length of the longest side of the triangle?

Rationale

The correct answer is 12. It is given that the triangle has angle measures of 30° , 60° , and 90° , and so the triangle is a special right triangle. The side measures of this type of special triangle are in the ratio $2:1:\sqrt{3}$. If x is the measure of the shortest leg, then the measure of the other leg is $\sqrt{3}x$ and the measure of the hypotenuse is $2x$. The perimeter of the triangle is given to be $18 + 6\sqrt{3}$, and so the equation for the perimeter can be written as $2x + x + \sqrt{3}x = 18 + 6\sqrt{3}$. Combining like terms and factoring out a common factor of x on the left-hand side of the equation gives $(3 + \sqrt{3})x = 18 + 6\sqrt{3}$. Rewriting the right-hand side of the equation by factoring out 6 gives $(3 + \sqrt{3})x = 6(3 + \sqrt{3})$. Dividing both sides of the equation by the common factor $(3 + \sqrt{3})$ gives $x = 6$. The longest side of the right triangle, the hypotenuse, has a length of $2x$, or $2(6)$, which is 12.

Question ID: 310c87fe

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

Answer

- A. 18
- B. 27
- C. 36
- D. 81

Correct Answer: B

Rationale

Choice B is correct. The surface area of a cube with side length s is equal to $6s^2$. Since the surface area is given as 54 square meters, the equation $54 = 6s^2$ can be used to solve for s . Dividing both sides of the equation by 6 yields $9 = s^2$. Taking the square root of both sides of this equation yields $3 = s$ and $-3 = s$. Since the side length of a cube must be a positive value, $s = -3$ can be discarded as a possible solution, leaving $s = 3$. The volume of a cube with side length s is equal to s^3 . Therefore, the volume of this cube, in cubic meters, is 3^3 , or 27.

Choices A, C, and D are incorrect and may result from calculation errors.